BMC Pulmonary Medicine

https://doi.org/10.1186/s12890-025-04045-6

Article in Press

Can assessment of small airway dysfunction support the diagnosis or management of asthma exacerbations in children? A systematic review

Received: 17 September 2025

Accepted: 24 November 2025

Published online: 09 December 2025

Cite this article as: Alshehri A., Alshahrani M.I., Sapey E. *et al.* Can assessment of small airway dysfunction support the diagnosis or management of asthma exacerbations in children? A systematic review. *BMC Pulm Med* (2025). https://doi.org/10.1186/s12890-025-04045-6

Abdulrahman Alshehri, Mohammed Ibrahim Alshahrani, Elizabeth Sapey, Robert Stockley & Mohammed Almeshari

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

ARTICLE IN PRESS

Can assessment of small airway dysfunction support the diagnosis or management of asthma exacerbations in children? – a systematic review

Authors and Affiliations

1. Abdulrahman Alshehri*

School of Infection, Inflammation and Immunology, University of Birmingham, B15 2TT, UK. asa274@student.bham.ac.uk. Health Rehabilitation Sciences Department, College of Applied Medical Sciences, King Saud University, Riyadh 11421, Saudi Arabia. ORCiD ID: https://orcid.org/0009-0005-4994-4357.

2. Mohammed Ibrahim Alshahrani

School of Infection, Inflammation and Immunology, University of Birmingham, B15 2TT, UK. Mxa1362@student.bham.ac.uk.

ORCiD ID: https://orcid.org/0009-0007-1475-3010.

3. Elizabeth Sapey

School of Medical Sciences, University of Birmingham, B15 2TT, UK. E.sapey@bham.ac.uk.

ORCiD ID: https://orcid.org/0000-0003-3454-5482

4. Robert Stockley

School of Infection, Inflammation and Immunology, University of Birmingham, B15 2TT, UK. r.a.stockley@bham.ac.uk

ORCiD ID: https://orcid.org/0000-0003-3726-1207

5. Mohammed Almeshari

Health Rehabilitation Sciences Department, College of Applied Medical Sciences, King Saud University, Riyadh 11421, Saudi Arabia. malmeshari@ksu.edu.sa

ORCiD ID: https://orcid.org/0000-0001-8449-9491

*Corresponding author

ARTICLE IN PRESS

ARTICLE IN PRESS

Abstract

Background

Asthma exacerbations are acute episodes associated with worsening symptoms and lung function decline. In children, diagnosis and monitoring rely largely on clinical judgment and measures of large airway function, which may overlook peripheral airway involvement. Tests of small airway function, such as forced expiratory flow between 25% and 75% of vital capacity (FEF25–75) and impulse oscillometry (IOS), may offer additional physiological insights. The aim of this systematic review is to evaluate the evidence supporting the use of small airway tests during pediatric asthma exacerbations.

Main body

The protocol was registered on PROSPERO, and this systematic review followed established methodology. Electronic databases searched included MEDLINE (Ovid), EMBASE (Ovid), CINAHL (EBSCOhost), and CENTRAL (Cochrane Library). The search strategy combined subject headings and keywords relating to asthma exacerbations and small airway function (e.g., ‘asthma exacerbation’, ‘small airway dysfunction’, ‘impulse oscillometry’). Eligible studies included observational studies and randomised controlled trials assessing small airway tests during paediatric asthma exacerbations (aged <18 years). Risk of bias was assessed using appropriate validated tools according to study design.

Thirty-five studies met inclusion criteria. Thirty-one studies reported FEF25–75; four studies reported IOS parameters, including one that also included multiple-breath washout (MBW). Most studies found that small airway indices were impaired during exacerbations and improved after treatment. In many cases, FEF25–75 and IOS

ARTICLE IN PRESS

parameters (R5–R20, AX) showed greater relative change than forced expiratory volume in one second (FEV1), and small airways dysfunction persisted longer despite clinical recovery. All IOS studies achieved high feasibility in acute settings. No study evaluated whether small airway tests could be used to direct clinical management.

Conclusion

Physiological tests of small airway function appear feasible during acute paediatric asthma exacerbations and may detect abnormalities not captured by FEV1. However, no included studies evaluated whether incorporating these indices altered management or improved outcomes. While these tests show promise for future diagnostic or monitoring use, further validation in larger cohorts is needed before routine implementation.

Systematic Review Registration:

PROSPERO 2025 CRD42025623062. Available from: https://www.crd.york.ac.uk/PROSPERO/view/CRD42025623062.

Keywords

Asthma, paediatric asthma, children, asthma exacerbation, small airway dysfunction, FEF25–75, impulse oscillometry, multiple-breath washout.

ARTICLE IN PRESS

Background

Asthma is a heterogeneous chronic respiratory disease, characterized by chronic airway inflammation, variable airflow limitation, and often associated with airway hyperresponsiveness and structural remodeling [1]. In children, phenotypes such as viral-induced wheeze, often indistinguishable from early asthma in preschoolers, and

ARTICLE IN PRESS

atopic asthma contribute to diagnostic uncertainty, particularly in children under five years of age [2]. Asthma remains one of the most prevalent chronic diseases of childhood, with international studies showing wide variability in prevalence, symptom burden, and disease severity across regions [3]. Globally, an estimated 334 million people are affected by asthma, with children accounting for a substantial proportion of asthma-related morbidity, especially in low- and middle-income countries [3, 4]. While many children experience mild or intermittent symptoms, a subset suffer recurrent or severe exacerbations that impair lung development, reduce quality of life, and may lead to long-term airway remodeling [5, 6].

While asthma is typically diagnosed based on a characteristic clinical symptom pattern, objective testing is essential for confirming the diagnosis, assessing disease severity, and monitoring asthma control [1]. However, no single gold standard test has been universally accepted for pediatric asthma diagnosis, particularly in younger children or during acute illness [7, 8]. These challenges are compounded by the episodic nature of symptoms and the limited feasibility of conventional lung function tests in early childhood [9].

ARTICLE IN PRESS

Asthma exacerbations are episodes of progressive increase in symptoms such as wheeze, shortness of breath, chest tightness, or cough, usually accompanied by reduced lung function and clinical deterioration [1]. These events are commonly assessed using tools like the pediatric Respiratory Assessment Measure (PRAM), which standardizes severity scoring based on parameters such as wheezing intensity, chest retractions, and oxygen saturation [10]. Exacerbations are among the leading causes of emergency visits and hospitalizations in children with asthma [3, 4]. Pulmonary function assessment typically includes peak expiratory flow (PEF), forced expiratory volume in one second (FEV1), and the FEV1/FVC ratio, though these measures often

ARTICLE IN PRESS

correlate poorly with clinical severity and may miss early or subtle airway changes [11, 12].

This diagnostic limitation may partly reflect the contribution of small airway dysfunction (SAD) to asthma, which is increasingly recognized as a key component of asthma pathophysiology [7, 13]. The small airways, defined as bronchioles less than 2 mm in internal diameter, are located in the distal bronchial tree and might be disproportionately involved in asthma, despite appearing normal on conventional spirometry [13]. SAD has been observed even in asymptomatic children or those with normal FEV1, and may be particularly relevant in pediatrics due to smaller airway caliber and ongoing lung development [12, 14]. Undetected SAD may drive silent clinical deterioration, increased exacerbation risk, or poor treatment response. If left unresolved, it may contribute to airway remodeling and reduced lung function trajectories in later life [6, 15]. Studies using impulse oscillometry (IOS) have shown that abnormalities in peripheral airway resistance and reactance may exist even when traditional spirometric indices are within normal limits, and may correlate with asthma control more closely than FEV1 [16].

While spirometry remains the most widely used tool in asthma management, its sensitivity to peripheral airway involvement is limited [11]. Flow indices such as forced expiratory flow between 25 and 75 percent of vital capacity (FEF25–75) are sometimes used to infer small airway involvement, but are effort-dependent and show high variability [17]. Advanced techniques such as IOS, the forced oscillation technique (FOT), and multiple-breath washout (MBW) allow direct assessment of small airway physiology. These methods rely on tidal breathing and are therefore more feasible in younger children and during acute illness [14, 16]. For example, FOT has been used to detect bronchodilator response in preschool children, while MBW-derived indices such

ARTICLE IN PRESS

as the lung clearance index (LCI), a measure of ventilation inhomogeneity, offer sensitive detection of SAD [14].

Although these tools have shown promise in stable pediatric asthma and exacerbations in adult asthma, their role during acute exacerbations remains under active investigation [18]. To date, no systematic review has specifically addressed the role of small airway function testing in acute pediatric asthma exacerbations. If validated through future studies, small airway assessments could inform earlier intervention and support more personalized management approaches during acute illness.

While small airway dysfunction (SAD) has been increasingly recognized in both adult and pediatric asthma, no previous systematic review has comprehensively evaluated its role during acute exacerbations in children. Existing reviews and primary studies have largely focused on stable disease or adult populations, leaving uncertainty regarding feasibility, measurement reliability, and clinical utility in pediatric settings [18]. This review complements recent adult-focused evidence by uniquely synthesizing studies performed specifically during asthma exacerbations in children, integrating findings across spirometry-derived indices, oscillometry, and washout techniques to identify patterns relevant to clinical translation.

ARTICLE IN PRESS

This systematic review aimed to evaluate the published evidence on the use of small airway function tests to diagnose, monitor, or manage asthma exacerbations in children. It examined whether these techniques provide clinically meaningful information that could improve decision-making during pediatric asthma exacerbations.

ARTICLE IN PRESS

Main text

Materials and Methods

This systematic review was registered in PROSPERO (CRD42025623062) [19] and conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [20]. A completed PRISMA checklist is available in Additional file 1. Where outcome measures were sufficiently consistent across studies, meta-analysis was considered. Otherwise, a structured narrative synthesis was performed, and relevant findings were summarized using visual representation.

Following a preliminary scoping of the literature and expert consultation, small airway function tests were selected for inclusion based on their reported use in pediatric asthma: impulse oscillometry (IOS), forced oscillation technique (FOT), multiple-breath washout (MBW), and spirometry-derived parameters including forced expiratory flow between 25% and 75% of vital capacity (FEF25–75) [11, 14, 21]. These tests are considered among the most relevant and feasible tools for assessing small airway dysfunction in children due to their ability to detect peripheral airway abnormalities and their suitability for use during tidal breathing [11, 14]. Other indices such as forced expiratory volume in three seconds (FEV3) and six seconds (FEV6) were excluded due to limited use in studies [22]. The inclusion criteria were based on a predefined PICO framework, provided in Additional file 2.

ARTICLE IN PRESS

Eligibility Criteria

Studies were selected according to predefined inclusion and exclusion criteria:

ARTICLE IN PRESS

Inclusion criteria

- Children aged < 18 years with a diagnosis of asthma or acute asthma exacerbation.
- Assessment of small airway function during an exacerbation using one or more of the following physiological tests:
  - Forced oscillation technique (FOT) or impulse oscillometry (IOS)
  - Multiple-breath washout (MBW)
  - Spirometry-derived parameter forced expiratory flow between 25 % and 75 % of vital capacity (FEF25–75)
- No restriction on asthma severity.
- Studies that reported FEV_1 and/or PEF as comparators, reflecting current clinical standards.
- Study designs including randomized controlled trials, cohort, case–control, or cross-sectional studies

Exclusion criteria

- Studies that did not include pediatric participants, or where pediatric data could not be separated from adult data.
- Studies that did not measure small airway function during an exacerbation.
- Studies focused primarily on non-asthma respiratory conditions (e.g., bronchiolitis or cystic fibrosis).
- Case reports with fewer than 10 patients, review articles, editorials, and background papers.
- Studies where small airway indices were measured only during stable disease or follow-up.

ARTICLE IN PRESS

ARTICLE IN PRESS

Searches were conducted in December 2024 across four electronic databases: Ovid Embase, Ovid MEDLINE, EBSCO CINAHL, and the Cochrane Central Register of Controlled Trials (CENTRAL). The search strategy combined subject headings and keywords related to small airway function tests (IOS, FOT, FEF, MMEF, MBW), asthma, and exacerbation. There were no restrictions on language or publication date. The full search strategy is provided in Additional file 3.

Study selection

Search results were imported into EndNote 20 (Clarivate Analytics), where duplicates were removed. The records were then uploaded to Covidence, a web-based platform designed to streamline systematic review workflows, where title and abstract screening was conducted.

Titles and abstracts were screened by two independent reviewers (AA and ES) using the prespecified inclusion and exclusion criteria. Disagreements were resolved through discussion or, if necessary, by consulting a third reviewer (MA). The same process was applied to full-text screening.

ARTICLE IN PRESS

Outcomes

The primary outcome was the assessment of small airway function during hospitalized asthma exacerbations in children using one or more of the following physiological tests: forced expiratory flow between 25% and 75% of vital capacity (FEF25–75), also referred to as maximal mid-expiratory flow (MMEF); impulse oscillometry (IOS); forced oscillation technique (FOT); and multiple-breath washout (MBW) indices, including lung clearance index (LCI), acinar ventilation heterogeneity

ARTICLE IN PRESS

(Sacin), and conductive ventilation heterogeneity (Scond). To be eligible, measurements had to be performed during the exacerbation itself, either at initial presentation or during the hospital stay, and not solely after recovery.

Secondary outcomes included comparisons between small airway indices and conventional spirometry measures (FEV1 and peak expiratory flow), changes in small airway function during recovery, and correlations with clinical severity, validated symptom scores, or inflammatory biomarkers such as blood or sputum eosinophils.

Data extraction

For each included study, data were extracted using a customized electronic form developed and approved by all reviewers prior to the extraction phase. Extracted variables included study characteristics (design, year, authorship, and geographical location), patient characteristics (age, sample size, and inclusion and exclusion criteria), timing of measurements, and intervention details (small airway test used, device if reported, and measurement values during the exacerbation in absolute terms, percentage predicted, and % change where applicable). Comparator data, including FEV1 and peak expiratory flow, were extracted when reported in the same formats.

ARTICLE IN PRESS

Data extraction was conducted by one reviewer (AA) and independently verified by a second (ES). Discrepancies were resolved through discussion, with a third reviewer (MA) consulted if agreement could not be reached. Where data were missing or unclear, study authors were contacted for clarification.

Risk of bias and quality assessment

ARTICLE IN PRESS

The risk of bias was assessed using validated tools based on study design. Randomised controlled trials were assessed using the revised Cochrane risk-of-bias tool for randomised trials, RoB 2.0 [23]. Non-randomised interventional studies were assessed using the Risk of Bias In Non-randomised Studies of Interventions, ROBINS-I [24]. Diagnostic accuracy studies were assessed with the QUADAS-2 tool [25]. Observational studies were assessed using the Newcastle–Ottawa Scale, NOS [26]. A modified version of the NOS was applied to cross-sectional studies.

Two reviewers independently assessed each included study (AA and ES). Disagreements were resolved through discussion, and where necessary, a third reviewer (MA) was consulted to reach consensus.

Strategy for data synthesis

ARTICLE IN PRESS

Due to anticipated heterogeneity in outcome definitions, asthma classifications, measurement protocols, and reporting formats, a structured narrative synthesis was planned. Studies were to be grouped by test modality (FEF25–75, IOS/FOT, or MBW), and results summarized according to changes in small airway indices during exacerbations and their response to treatment, including comparisons with conventional spirometric parameters such as FEV1. Where appropriate, findings were presented visually to illustrate pre- and post-treatment changes. Meta-analysis was considered for outcomes reported consistently across studies; however, pooling was ultimately not pursued due to substantial methodological variability.

Results

Study selection

A total of 3006 articles were identified through database searching. After the removal of 762 duplicates, 2244 were screened by title and abstract. Sixty-six full-text articles were assessed for eligibility, and 35 studies met the inclusion criteria and were included in the final review. The study selection process is summarized in the PRISMA flow diagram (Figure 1). Articles excluded in the full-text screening phase are listed with reasons given in the Additional file 4.

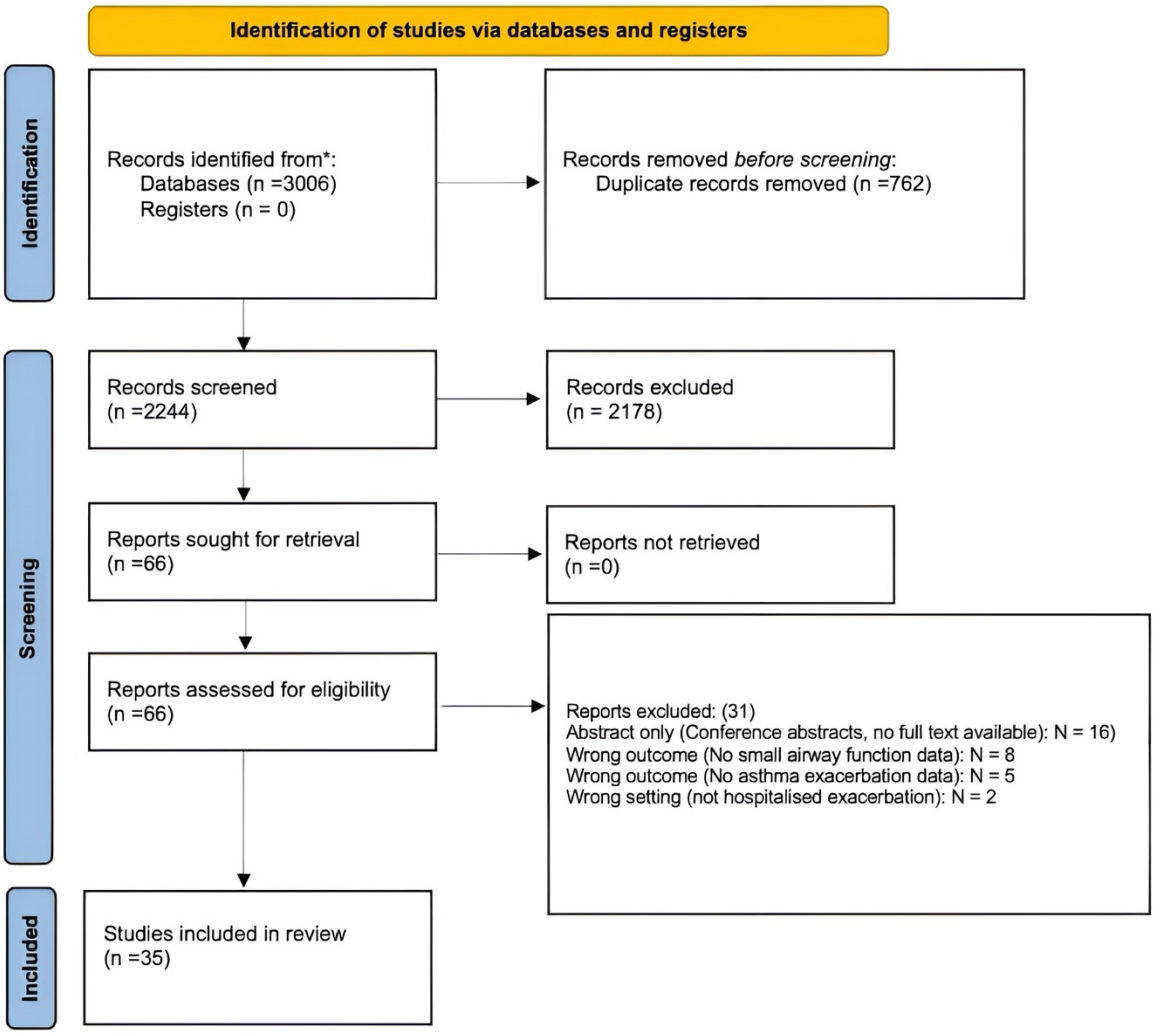


Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart of included studies.

ARTICLE IN PRESS

Study characteristics

This systematic review included 35 studies conducted across 10 countries and five continents: Asia, Europe, North America, South America, and Africa. The most frequently represented countries were China, Turkey, and Brazil, with additional studies from the United States, Canada, India, Italy, South Africa, Poland, and Taiwan. Most studies were single-center investigations conducted in emergency departments or inpatient hospital settings.

In terms of study design, 12 studies were observational, including both cross-sectional and prospective cohort designs; 16 were randomised controlled trials (RCTs), primarily evaluating bronchodilator or pharmacologic therapies during acute asthma exacerbations; 6 were non-randomised interventional studies; and 1 study used a diagnostic test accuracy (DTA) design.

ARTICLE IN PRESS

Across all studies, a total of 1,854 pediatric patients experiencing asthma exacerbations were included. Comparator groups were enrolled in several studies and comprised 316 patients with stable, chronic, or cough variant asthma, and 330 healthy control subjects. However, the depth and consistency of comparator data varied between studies.

Small airway function was assessed using a range of methods, including forced expiratory flow at 25–75% of vital capacity (FEF25–75), maximal mid-expiratory flow (MMEF), impulse oscillometry (IOS), the forced oscillation technique (FOT), and indices derived from multiple-breath washout (MBW), such as the lung clearance index (LCI), Sacin, and Scond. Measurements were reported variously as absolute values (liters per second), % predicted, or % change from baseline. Most studies treated these tests as

ARTICLE IN PRESS

secondary outcomes, and only a minority reported the specific devices or measurement protocols used.

The timing of small airway assessments varied widely. Some studies conducted testing solely during the acute phase of exacerbation, while others included follow-up measurements post-recovery. The definitions and criteria for confirming an asthma exacerbation were also heterogeneous. While some studies followed formal guidelines, such as those of the Global Initiative for Asthma (GINA) or the American Thoracic Society (ATS), others relied on prior asthma diagnosis or clinical judgment without specifying formal criteria.

Due to substantial methodological heterogeneity across studies (including differences in design, test timing, outcome definitions, and analytic approaches) a meta-analysis was not appropriate. Instead, a narrative synthesis approach was adopted. Table 1 summarizes the key characteristics of all included studies.

ARTICLE IN PRESS

Table 1. Characteristics of Included Studies

Study ID	Sample (n)	Severity Classification	Age (years)	small airway function Baseline	small airway function follow-up	FEV1	PEF	Diagnosis criteria
Nakanishi et al 1997[27] USA Randomised controlled trial	113 asthma exacerbation Ultrasonic Nebulizer (UN) N: 67 vs Jet Nebulizer (JN) N:46	Moderately severe asthma	UN:10.1 ± 3.3; JN: 11.0 ± 3.1	NR	NR	UN: 36.1 ± 18.5%; JN: 38.1± 16.9%	NR	Previous diagnosis of asthma
Özdemir et al 2020[28] Turkey non-randomised	115 children with acute asthma; Mild asthma (n = 50)	Mild (FEV1 60–75%), Moderate (FEV1 40–59%)	Mild: 10.30 ± 2.91; Moderate: 10.54 ± 3.11	Mild: 58.20 ± 12.24%; Moderate: 37.77 ± 11.37%	Mild: 66.57 ± 16.95%; Moderate: 43.41 ± 14.19%	Mild: 67.14 ± 4.99%; Moderate: 48.50 ± 6.81%	Mild: 65.49 ± 12.32%; Moderate: 47.56 ± 11.78%	Previous diagnosis of asthma

ARTICLE IN PRESS

interventional study	Moderate asthma (n = 65)							
Becker et al 1983* [29] Canada Randomised controlled trial	Acute asthma 40 (20 salbutamol, 20 epinephrine)	Not Specified	Salbutamol: 10.6 ± 0.7; Epinephrine: 10.4 ± 0.7	Reported in figure only	Reported in figure only	Reported in figure only	NR	Previous diagnosis of asthma
Chakraborti et al 2006[30] India Randomised controlled trial	Asthma exacerbations 60 (30 salbutamol & ipratropium, 30 salbutamol & placebo)	Mild to moderate exacerbation	Group 1: 106 ± 25.8 months; Group 2: 118 ± 25.2 months	Group 1: 80.1 ± 65.5%; Group 2: 84.5 ± 129.8%	Group 1: 121.0 ± 90.7%; Group 2: 72.2 ± 33.3%	Group 1: 72.3 ± 37.1%; Group 2: 57.4 ± 21.0%	Group 1: 64.36 ± 28.3%; Group 2: 57.3 ± 19.20%	Previous diagnosis of asthma
Garra et al 1997* [31] USA Randomised controlled trial	30 hospitalized children with acute asthma (15 received isoproterenol, 15 received metaproterenol)	Severe exacerbation requiring hospitalization	Isoproterenol: 10.0 ± 1.1; Metaproterenol: 10.1 ± 1.1	Isoproterenol: 15.3 ± 2.6%; Metaproterenol: 11.8 ± 1.6%	Reported in figure only	Isoproterenol: 30.3 ± 3.4%; Metaproterenol: 24.8 ± 4.4%	NR	Previous diagnosis of asthma
Caffarelli et al 2012[32] Italy non-randomised interventional study	12 healthy controls. 22 asthmatic children with acute asthma exacerbation.	Mild–moderate exacerbation	Healthy controls: 11.7 ± 2.8; Asthmatics: 9.4 ± 2.7	Healthy controls: 106.2 ± 19.8%. Asthmatics: 76.6 ± 18.3%	Asthmatics: 81.9 ± 19.3%	Healthy controls: 112.6 ± 15.8%; Asthmatics: 95.0 ± 11.9%	NR	Global Initiative for Asthma
Caffarelli et al 2014[33] Italy non-randomised interventional study	Healthy Control Group: 28 children Asthma Exacerbation Group: 12 children	Mild–moderate exacerbation	Healthy controls: 5.1 ± 1.4; Asthmatics: 9.3 ± 2.4	Asthmatics: 83.4 ± 12.8%	Asthmatics: 85.5 ± 22.17%	Asthmatics: 98.7 ± 0.66%	NR	Global Initiative for Asthma
Scalabrin et al 1993[34] Brazil Randomised controlled trial	35 episodes in 21 children; Oral salbutamol: 13, Open nebulization: 11, Closed-port nebulization: 11	Acute bronchial asthma	mean 10.1 years (range: 7–14 years)	OR:1.12 ± 0.44 L/s; ON:0.78 ± 0.34 L/s; CN: 0.79 ± 0.41 L/s	NR	OR: 55.27%; ON: 45.07%; CN: 42.44%	NR	Previous diagnosis of asthma
Scalabrin et al 1996[35]	37 episodes in 21	Acute bronchial asthma	Mean: 10.41 years	SAL: 0.79 ± 0.41; TER: 0.71	NR	SAL: 42.4%; TER:	NR	Previous diagnosis of asthma

Brazil Randomised controlled trial	children; groups: Salbutamol (11), Terbutaline (12), Fenoterol (14)		(range 7–14)	± 0.44; FEN: 1.01 ± 0.65		42.9%; FEN: 53.1%		
Sharma et al 2023[36] India Diagnostic test accuracy	50 children with acute asthma exacerbation.	Mild–moderate and severe exacerbation	9.66 ± 1.63 years	1.24 ± 0.55 L/s	1.56 ± 0.64 L/s	1.09 ± 0.32 L/S	2.40 ± 0.96 L/S	Global Initiative for Asthma
Lowenthal et al 1992[37] Randomised controlled trial	64 children with acute asthma; Facemask: 32, Mouthpiece: 32.	mild to moderate	Facemask: 13.5 ± 4.4; Mouthpiece: 13.0 ± 3.6	Facemask: 28.6 ± 14.3%; Mouthpiece: 31.4 ± 16.5%	NR	Facemask: 45.3 ± 16.3%; Mouthpiece: 50.7 ± 15.1%	Facemask: 53.4 ± 22.2%; Mouthpiece: 54.7 ± 20.0%	American Thoracic Society (ATS)
Sol et al 1999[38] Brazil Observational study	174 children with acute exacerbation (Allergy Clinic: 74, Emergency Room: 100)	mild to severe based on clinical score	AC: 8.6 y: ER: 4.6 y	AC group: 51.1%, ER: not reported	AC group: 70.8%, ER: not reported	AC group: 62.7%, ER: not reported	NR	Previous diagnosis of asthma
Uden et al 1985[39] USA Randomised controlled trial	Asthma exacerbation: total analyzed = 19 (Terbutaline: 7; Epinephrine: 12)	moderate severity	Mean: 11.2 ± 2.9 years (range: 5.8–17.5)	Reported in figure only	Terbutaline: 31.8 ± 18.6%; Epinephrine: 39.0 ± 12.2%	Reported in figure only	NR	Previous diagnosis of asthma
Carter et al 1996[40] USA Randomised controlled trial	11 children ACUTE ASTHMA	Acute severe asthma	Range: 7–18 years	30 ± 14%	32 ± 14%	49 ± 16%	48 ± 16%	American Thoracic Society (ATS)
Baraldi et al 1997* [41] Italy Non-randomised, controlled, interventional study	Total = 39 Acute asthma = 16 Healthy controls = 16 Stable asthma with albuterol challenge = 7	NR	Healthy controls: 9.3 ± 0.5 years; Stable asthma: 10.6 ± 0.7 years; Exacerbation group: 9.3 ± 0.5 years.	Healthy controls: 114.4 ± 4.9%; Asthmatics: 46.5 ± 5.1 %	Asthmatics: 93.3 ± 8.0%	Healthy controls: 93.8 ± 2.2 %; Asthmatics: 62.4 ± 4.4 %	NR	objective testing and clinical criteria.
Shapiro et al 1987* [42] USA Randomised	100 children (ages 6–12) with acute asthma	Mild to moderate acute exacerbation	Age: 6–12 years (mean not reported)	Placebo: 0.50 ± 0.05 L/s; Metaproterenol 5 mg: 0.53	Reported in figure only	Placebo: 0.77 ± 0.07 L; Metaproterenol 5 mg: 0.84	Placebo: 1.16 ± 0.10 L/s; Metaproterenol 5	Previous diagnosis of asthma

ARTICLE IN PRESS

controlled trial	exacerbation; randomised to saline (n=25), metaproterenol 5 mg (n=25), 10 mg (n=25), or 15 mg (n=25).			± 0.06 L/s; 10 mg: 0.72 ± 0.07 L/s; 15 mg: 0.74 ± 0.07 L/s		± 0.08 L; 10 mg: 0.99 ± 0.07 L; 15 mg: 1.02 ± 0.08 L	mg: 1.29 ± 0.12 L/s; 10 mg: 1.41 ± 0.10 L/s; 15 mg: 1.45 ± 0.11 L/s	
Zar et al 1999[43] South Africa Randomised controlled trial**	88 children with acute asthma. Arms: Conventional spacer (n=22), sealed bottle (n=22), unsealed bottle (n=22), polystyrene cup (n=22).	PEF 60–79% (mild), PEF 20–59% (moderate/severe)	Conventional spacer: 10.3 (8.9–12.4) years; Sealed bottle: 9.2 (7.9–10.7) years; Unsealed bottle: 9.1 (7.9–10.7) years; Cup: 9.7 (7.5–11.3) years.	Conventional spacer: 37 (27–52); Sealed bottle: 32 (24–47); Unsealed bottle: 40 (26–50); Cup: 37.5 (27–51).	NR	Conventional spacer: 67 (57–77); Sealed bottle: 67.5 (51–74); Unsealed bottle: 64.5 (52–75); Cup: 65 (53–77).	Conventional spacer: 58.5 (45–69); Sealed bottle: 59.5 (43–68); Unsealed bottle: 59 (32–73); Cup: 54.5 (44–67).	Previous diagnosis of asthma
Debley 2012[44] USA Observational study	Asthmatic children (exacerbation group): 34 completed follow-ups; Healthy controls: 40 enrolled.	Severe exacerbation	Healthy controls: 10.8 ± 3.2 years; Asthma (exacerbation): 9.7 ± 2.9 years.	Healthy controls: 90.7 ± 19.7%; Asthma (exacerbation): 43.3 ± 26.7%	Not reported for the whole cohort	Healthy controls: 99.3 ± 10.8%; Asthma (exacerbation): 76.9 ± 20.5% (FEV1).	NR	National Heart, Lung, and Blood Institute/ National Asthma Education and Prevention Program
Yuan et al 2013[45] China Observational study	Total patients: 170 Healthy controls: 30 Stable asthma: 50 Acute asthma exacerbation: 50 Cough-variant asthma (CVA): 40	Mild to moderate exacerbation	Healthy: 8.6 ± 2.1; Stable asthma: 8.0 ± 2.1; Asthma exacerbation: 8.3 ± 2.0; CVA: 7.9 ± 2.0	Healthy: 97 ± 12%; Stable asthma: 84 ± 18%; Asthma exacerbation: 50 ± 17%; CVA: 79 ± 22%	NR	Healthy: 107 ± 16 %; Stable asthma: 98 ± 14%; Asthma exacerbation: 69 ± 12%; CVA: 98 ± 14%	Healthy: 106 ± 14 %; Stable asthma: 95 ± 12%; Asthma exacerbation: 67 ± 14%; CVA: 93 ± 17%	Previous diagnosis of asthma
Malka et al 2015[46] USA Observational study	66 children ACUTE ASTHMA (during stable period &	Mild, moderate, and severe persistent asthma.	11 ± 3.2 years (Mean ± SD)	stable period (Mean ± SD): 76.6 ± 26.8% predicted;	71.0 ± 4.9% predicted (model-estimated mean	Stable period (Mean ± SD): 95.0 ± 16.0% predicted;	NR	Previous diagnosis of asthma

	exacerbation)			Exacerbation (Mean ± SEM): 43 ± 2% predicted	from Table III)	Exacerbation (Mean ± SEM): 72 ± 2% predicted		
Maria Wawszczak [47] **Poland Observational study**	42 children with asthma Groups: Stable asthma (n = 22), Exacerbation (n = 20)	Mixed	Age (median, IQR): Stable: 12 (7–17); Exacerbation: 12.5 (6–17)	R5–R20 (kPa·L^{-1}·s, median [IQR]): Stable: 0.12 (0.07–0.16); Exacerbation: 0.18 (0.13–0.295) MBW z-scores (median [IQR]): • Scond: Stable: 0.35 (–0.5 to 1.5); Exacerbation: 7.7 (4.6 to 8.9) • Sacin: Stable: 0.4 (–1.3 to 1.7); Exacerbation: 2.2 (1.6 to 3.2) • LCI: Stable: 1.6 (0.7 to 3.4); Exacerbation: 8.2 (5.85 to 11.4)	NR	(z-score, median [IQR]): Stable asthma: 0.0 (–1.37 to 0.59); Exacerbation: –1.98 (–3.01 to –1.42)	NR	Global Initiative for Asthma
La Rosa et al 1997[48] Italy Randomised controlled trial	28 children with acute asthma exacerbation: Salbutamol + Flunisolide (n=15), Salbutamol only (n=13)	Intermittent or mild persistent exacerbation	Age range: 5–14 years Mean age: 9.0 ± 2.5 years (whole cohort)	68.8 ± 7.6% (salbutamol), 66.9 ± 5.0% (salbutamol + flunisolide)	Day 15: 81.3 ± 5.5%(salbutamol), 97.1 ± 6.4% (salbutamol + flunisolide)	65.6 ± 2.8%(salbutamol), 65.8 ± 3.3% (salbutamol + flunisolide)	72.7 ± 4.7 %(salbutamol), 63.8 ± 3.8% (salbutamol + flunisolide)	Clinical and spirometry-based
Yilmaz et al 2014[49] Turkey Observational study	Total patients: 57 children with moderate to severe asthma exacerbation	GINA 2012 criteria (moderate/severe)	10.7 ± 2.4 years	48 ± 17%	Day 35: 90 ± 6%	67 ± 17%	63 ± 16%	Global Initiative for Asthma

ARTICLE IN PRESS

Comberiati et al 2020[50] USA Observational study	92 children with asthma exacerbation: 37 African American, 55 Non-African American (White and Hispanic).	Mild to moderate persistent asthma (NHLBI)	Overall cohort: Mean age = 11.0 ± 3.3 years	African American – Stable: 71.4 ± 35.6%; Non-African American – Stable: 75.0 ± 24.2%; African American – Exacerbation: 51.0 ± 24.6%; Non-African American – Exacerbation: 45.3 ± 17.7%	African American: 68.7 ± 33.6; Non-African American: 67.8 ± 21.9	African American – Stable: 86.3 ± 17.3%; Non-African American – Stable: 96.3 ± 14.2%	NR	Based on previous physician diagnosis and NAEPP treatment context
Chandra et al 2004† [51] India Randomised controlled trial	60 children with mild or moderate asthma exacerbation (Terbutaline: n = 31; Salbutamol: n = 29).	Mild and moderate (defined by symptoms and PEFR % predicted)	Terbutaline: Median age = 10 years (IQR 9–11); Salbutamol: Median age = 9 years (IQR 7.8–11)	Terbutaline: 63% (52.9–77); Salbutamol: 57% (52.7–66.3) (likely mislabeled as PEFR in Table 1)	Terbutaline: 86% (75.5–99); Salbutamol: 84% (67.5–98.1)	Terbutaline: 54% (45–62); Salbutamol: 52% (48–63.8) median (95% confidence intervals)	Terbutaline: 69% (59.5-75); Salbutamol: 72% (61.3-90.1)	Not reported
Cengizlier et al 1999[52] Turkey Randomised controlled trial	40 children with acute asthma exacerbation (Glucagon: n=20; Salbutamol: n=20)	NR	9.9 ± 2.1 (glucagon), 10.7 ± 2.1 (salbutamol)	Reported in figure only	Reported in figure only	Reported in figure only	Reported in figure only	American Thoracic Society (ATS)
Samransamruajkit et[53] al 2003 USA Randomised controlled trial	40 children with acute asthma exacerbation; Flutter therapy (n=20), Standard care (n=20)	Moderate asthma exacerbation	standard care: 10.82 ± 0.7; Flutter: 9.19 ± 0.42	standard care: 27.1 ± 4.1; Flutter: 20.6 ± 3.1	Reported in figure only	Standard care: 46.4 ± 3.5%; Flutter: 38.8 ± 3.0%	NR	Previous diagnosis of asthma
Batmaz et al 2015[54] Turkey Observational study	245 children total 35 with acute asthma exacerbation, 107 with stable asthma, and 103	Mild	Acute asthma: 10.82 ± 2.53 years; Stable asthma: 10.57 ± 2.5 years; Healthy	Acute asthma: FEF25–75 = 52.3 ± 28.4% predicted; R5–R20 = 0.35 ± 0.18 kPa·L^{-1}·s;	NR	Acute asthma: 83.5 ± 19.5%; Stable asthma: 95.7 ± 13.2%; Healthy controls:	NR	National Heart, Lung, and Blood Institute

ARTICLE IN PRESS

ARTICLE IN PRESS

	healthy controls		controls: 11.35 ± 2.54 years	Stable asthma: FEF25–75 = 70.6 ± 24.5% predicted; R5–R20 = 0.24 ± 0.10 kPa·L⁻¹·s; Healthy controls: FEF25–75 = 100.7 ± 27.9% predicted; R5–R20 = 0.18 ± 0.11 kPa·L⁻¹·s.		105.9 ± 13.9%		
Zanconato et al 2002* [55] Italy Observational study	30 children with acute asthma exacerbation; 29 healthy controls	NR	Age (Mean ± SEM): Asthma: 10.5 ± 0.5 years; Healthy controls: 8.5 ± 0.5 years	Asthma: 42%	Asthma: 66%	Asthma: 68%	NR	American Thoracic Society (ATS)
Lee et al 1998[56] Taiwan Observational study	34 children (22 with asthma + allergic rhinitis, 12 with allergic rhinitis only)	NR	8 years (range 3y3m to 14y)	Asthma exacerbation: 78.6 ± 26.7%	Asthma exacerbation: 92.8 ± 25.4%	Asthma exacerbation: 80.9 ± 19.7%	NR	NHLBI 1997 guidelines
Liu et al 2007[57] China Observational study	Acute asthma (exacerbation): 40 patients; Remission (same cohort post-treatment): 23 patients; Healthy controls: 25 patients	Mild-to-moderate asthma exacerbation	Healthy controls: 4.9 ± 0.2 years; Acute asthma: 4.1 ± 0.3 years; Asthma remission (acute asthma after treatment): 4.5 ± 0.8 years	R5–R20 (kPa·L⁻¹·s, Healthy controls: 0.24 ± 0.06; Acute asthma: 0.53 ± 0.24;	R5–R20 (kPa·L⁻¹·s, Asthma remission (acute asthma after treatment): 0.35 ± 0.11	NR	NR	Chinese national guideline for pediatric asthma
Nuhoğlu et al 2001[58] Turkey Randomised controlled trial	Total: 60 children with acute asthma exacerbation Group I (high-dose inhaled	Mild-to-moderate acute asthma exacerbations	age: 8.91 ± 2.60 years; age: 9.54 ± 3.05 years	Group I: Baseline: 62.42 ± 25.50; Group II: Baseline: 56.17 ± 25.18	Group I: Post-treatment: 84.23 ± 29.44; Group II: Post-treatment	Group I: Baseline: 78.61 ± 18.23; Group II: Baseline: 72.38 ± 17.03	NR	American Thoracic Society (ATS)

ARTICLE IN PRESS

	budesonide): n=31; Group II (oral methylprednisolone + medium-dose budesonide): n=29				: 74.03 ± 24.71			
Lin et al 1995[59] Taiwan Randomised controlled trial	Total: 111 children with acute asthma exacerbation MDI + spacer (n=56), nebulized terbutaline (n=55)	Mild-to-moderate acute asthma	MDI Group n = 56, Age: 8.1 ± 2.6 years; Nebulizer Group n = 55, Age: 8.4 ± 3.0 years	Group 1 (MDI): Baseline: 57.53 ± 28.02; Group 2 (Nebulizer): Baseline: 55.63 ± 28.77	Group 1 (MDI): 75.24 ± 31.25; Group 2 (Nebulizer): 63.82 ± 28.83	Group 1 (MDI): Baseline: 65.74 ± 20.33; Group 2 (Nebulizer): Baseline: 63.14 ± 20.50	Group 1 (MDI): Baseline: 57.56 ± 19.60%; Group 2 (Nebulizer): Baseline: 60.30 ± 21.32%	American Thoracic Society (ATS)
Lin et al 1996*[60] Taiwan non-randomised interventional study	90 children with acute asthma exacerbation (Terbutaline nebulization: n = 45; Epinephrine injection: n = 45)	Mild, moderate, and severe based on FEV1 and clinical scoring	Group 1 (Terbutaline): n = 45, age: 7.78 ± 0.35 years; Group 2 (Epinephrine): n = 45, age: 7.47 ± 0.31 years	Group 1 (Terbutaline): Baseline: 59.74 ± 4.33; Group 2 (Epinephrine): Baseline: 65.73 ± 4.39	Group 1 (Terbutaline): Post-treatment: 65.34 ± 4.39; Group 2 (Epinephrine): Post-treatment: 80.36 ± 5.04	Group 1 (Terbutaline): Baseline: 65.45 ± 3.08; Group 2 (Epinephrine): Baseline: 65.81 ± 3.05	NR	American Thoracic Society (ATS)
Yu et al 2011[61] China Observational study**	166 preschool children (age 3–6 years): Acute asthma (n=44); non-acute asthma (n=41); Cough variant asthma (n=46); Healthy controls (n=35)	Mild-to-moderate	(reported range: 3–6 years; group-specific values not separately reported)	R5–R20 (kPa·L^{-1}·s, Median [IQR]): Healthy controls: 0.08 (0.04–0.17); Acute asthma: 0.20 (0.15–0.28); non-acute asthma: 0.13 (0.10–0.19); Cough variant asthma: 0.12 (0.08–0.16)	NR	NR	NR	Chinese national and international diagnostic standards

Legend: Values reported in mean (SD)
*Values reported in mean (SEM)
**Values reported in Median (interquartile range, 25th–75th percentile)
† Median with 95% confidence intervals

ARTICLE IN PRESS

ID = identification, NR = Not Reported.

Risk of Bias and Quality Assessment

ARTICLE IN PRESS

Risk of bias was assessed using four validated tools, selected according to study design. Twelve observational studies were evaluated using the Newcastle–Ottawa Scale (NOS) [26] (Figure 2A). While some studies performed well in domains such as outcome ascertainment and adequacy of follow-up, most were limited by unclear cohort representativeness, lack of group comparability, or incomplete reporting of selection procedures. These limitations introduced varying degrees of bias, particularly in the selection and comparability domains, which may affect the internal and external validity of the findings.

ARTICLE IN PRESS

Sixteen randomised controlled trials were assessed using the revised Cochrane risk-of-bias tool for randomised trials (RoB 2.0) [23] (Figure 2B). Several trials raised “some concerns,” primarily due to insufficient detail on randomization procedures or deviations from intended interventions. A subset of trials was judged to be at high risk of bias due to missing outcome data or inappropriate handling of the analysis population.

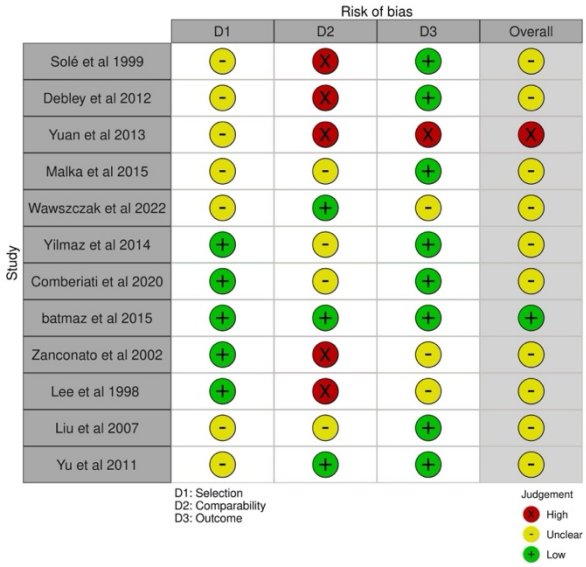
Six non-randomised interventional studies were assessed using the Risk of Bias in Non-randomised Studies of Interventions (ROBINS-I) tool [24] (Figure 2C). These studies frequently exhibited serious risk of bias in domains related to confounding, classification of interventions, and outcome measurement. Inconsistent reporting of selection criteria and outcome definitions contributed to elevated risk across multiple domains.

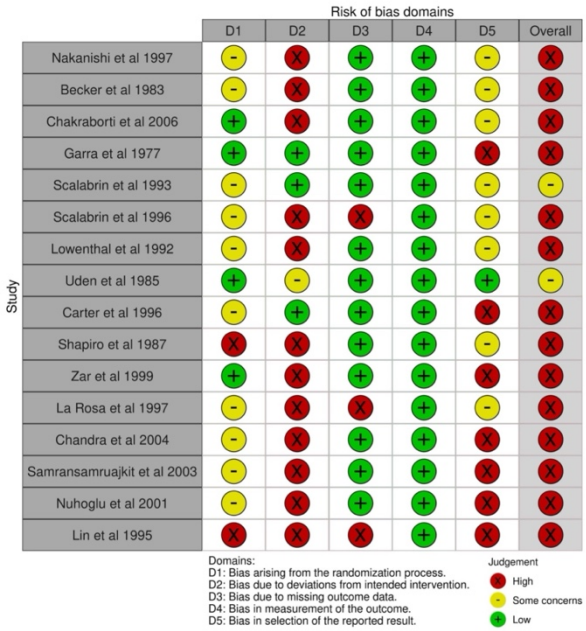
One diagnostic test accuracy (DTA) study was evaluated using the QUADAS-2 tool [25] (Figure 2D). The risk of bias was generally low, although some concerns remained regarding patient flow and timing.

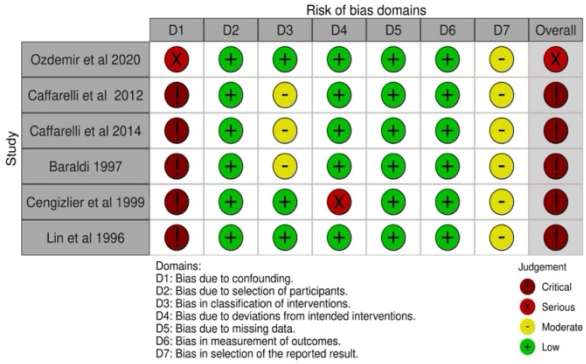
In several studies, small airway indices such as FEF25–75 were analyzed as secondary or exploratory outcomes rather than predefined endpoints, increasing the potential for selective reporting bias. This limitation was considered when assigning overall risk-of-bias ratings, particularly within the outcome measurement domain [23-26].

Risk-of-bias plots were generated using the robvis tool to visualize domain-level and overall risk of bias across studies [62].

ARTICLE IN PRESS


Risk of bias
D1
D2
D3
Overall
Study
Solé et al 1999
Debley et al 2012
Yuan et al 2013
Malka et al 2015
Wawszczak et al 2022
Yilmaz et al 2014
Comberiati et al 2020
batmaz et al 2015
Zanconato et al 2002
Lee et al 1998
Liu et al 2007
Yu et al 2011
D1: Selection
D2: Comparability
D3: Outcome
Judgement
High
Unclear
Low


Risk of bias domains
D1
D2
D3
D4
D5
Overall
Study
Nakanishi et al 1997
Becker et al 1983
Chakraborti et al 2006
Garra et al 1977
Scalabrin et al 1993
Scalabrin et al 1996
Lowenthal et al 1992
Uden et al 1985
Carter et al 1996
Shapiro et al 1987
Zar et al 1999
La Rosa et al 1997
Chandra et al 2004
Samransamruajkit et al 2003
Nuhoglu et al 2001
Lin et al 1995
Domains:
D1: Bias arising from the randomization process.
D2: Bias due to deviations from intended intervention.
D3: Bias due to missing outcome data.
D4: Bias in measurement of the outcome.
D5: Bias in selection of the reported result.
Judgement
High
Some concerns
Low


Risk of bias domains
D1
D2
D3
D4
D5
D6
D7
Overall
Study
Ozdemir et al 2020
Caffarelli et al 2012
Caffarelli et al 2014
Baraldi 1997
Cengizlier et al 1999
Lin et al 1996
Domains:
D1: Bias due to confounding.
D2: Bias due to selection of participants.
D3: Bias in classification of interventions.
D4: Bias due to deviations from intended interventions.
D5: Bias due to missing data.
D6: Bias in measurement of outcomes.
D7: Bias in selection of the reported result.
Judgement
Critical
Serious
Moderate
Low

Risk of bias domains
D1
D2
D3
D4
Overall
Study
Sharma et al 2023
Domains:
D1: Patient selection.
D2: Index test.
D3: Reference standard.
D4: Flow & timing.
Judgement
Low

371

ARTICLE IN PRESS

Figure 2. Risk of bias assessments. (A) Observational studies assessed using the Newcastle–Ottawa Scale (NOS) [26]. (B) Randomised controlled trials assessed using the revised Cochrane RoB 2.0 tool [23]. (C) Non-randomised interventional studies assessed using the ROBINS-I tool [24]. (D) Diagnostic test accuracy study assessed using the QUADAS-2 tool [25].

Results of individual studies

The 35 included studies were grouped based on the small airway test reported: forced expiratory flow between 25% and 75% of vital capacity (FEF25–75) or indices derived from impulse oscillometry (IOS) and the forced oscillation technique (FOT). A total of 31 studies assessed FEF25–75 using spirometry, while 4 reported IOS/FOT parameters such as R5–R20 and AX.

Individual findings are summarized below, with emphasis on the direction and magnitude of small airway responses and how they compare to conventional spirometric indices such as FEV1.

ARTICLE IN PRESS

Forced expiratory flow between 25% and 75% of vital capacity (FEF25–75)

Across the 31 included studies that assessed FEF25–75, most reported statistically significant improvement following treatment with bronchodilators or corticosteroids, often with numerically larger % changes than FEV1. In several trials, FEF25–75 appeared more sensitive to treatment effects or more delayed in recovery, suggesting possible peripheral airway involvement not fully captured by conventional indices. A few studies found that FEF25–75 remained abnormal even after symptom resolution or FEV1 normalization. However, small airway function was rarely analyzed

ARTICLE IN PRESS

independently, and in many cases, FEF25–75 results were reported descriptively without formal statistical testing or confounder adjustment.

Studies with a greater response in FEF25–75 compared with FEV1

Overall, 16 out of 35 studies demonstrated a greater numerical or statistically significant improvement in FEF25–75 compared with FEV1 during acute asthma exacerbations or recovery. Nakanishi et al. [27] compared ultrasonic versus jet nebulization in 113 children with acute asthma. Both FEV1 and FEF25–75 improved, with FEF25–75 showing numerically greater % change (ultrasonic: 33.2% vs. 24.0%; jet: 31.2% vs. 23.8%), though the difference was not statistically significant.

Özdemir et al. [28] conducted a non-randomised interventional study in Turkey evaluating intravenous magnesium sulfate in 115 children with acute asthma. Spirometry was performed before and after a single infusion, with FEV1 and FEF25–75 both showing statistically significant improvement. In the mild exacerbation subgroup, FEV1 increased from 67.14% to 72.29%, and FEF25–75 increased from 58.20% to 66.57%. In the moderate subgroup, FEV1 improved from 48.50% to 53.78%, and FEF25–75 rose from 37.77% to 43.41%. While small airway dysfunction was not the focus of the analysis, the consistent FEF25–75 response suggests potential small airway involvement.

Chakraborti et al. [30] conducted a double-blind randomised controlled trial comparing salbutamol alone to salbutamol plus ipratropium in 60 children with mild to moderate acute asthma. Spirometry was performed before and 30 minutes after treatment. Both FEV1 and FEF25–75 (L/s) improved in both groups, with greater gains in the combination therapy arm. Post-treatment FEF25–75 (L/s) was significantly higher

ARTICLE IN PRESS

ARTICLE IN PRESS

in the salbutamol + ipratropium group than in the salbutamol-only group (2.25 ± 1.10 vs. 1.39 ± 0.71 L/s, p = 0.01), while FEV1 (L) showed a non-significant trend toward higher values in the combination group (1.51 ± 0.80 vs. 1.20 ± 0.63 L, p = 0.06).

Caffarelli et al. [32] evaluated exhaled hydrogen peroxide (H_2O_2) and lung function in 22 children during acute asthma exacerbation and one week after treatment. The study did not explicitly investigate small airway dysfunction. While clinical symptoms improved, spirometry showed only modest changes: FEV1 increased from 94.9% ± 11.9 to 101.5% ± 13.9 predicted (p = 0.135), while FEF25–75 increased from 76.6% ± 18.3 to 81.9% ± 19.3 predicted (p = 0.001).

Solé et al. [38] evaluated 174 children with acute asthma or wheezing in an observational study, comparing clinical scores, pulse oximetry (SpO_2), and spirometric parameters. In the subset able to perform spirometry, both FEV1 and FEF25–75 improved after bronchodilator treatment. FEF25–75 values were significantly correlated with SpO_2, though small airway function was not explicitly analyzed.

ARTICLE IN PRESS

Baraldi et al. [41] conducted a non-randomised controlled interventional study assessing exhaled nitric oxide (NO) and spirometry in 16 children with acute asthma before and after 5 days of oral prednisone. FEF25–75 improved post-treatment, though values remained lower than in healthy controls.

Shapiro et al. [42] conducted a double-blind, randomised dose–response trial comparing 0, 5, 10, and 15 mg of nebulized metaproterenol in 100 children with acute asthma. Spirometry was assessed at multiple time points, and FEF25–75 was among the reported outcomes. The largest FEF25–75 improvement occurred with 5 mg metaproterenol, with diminished response at 15 mg. While FEV1 also improved, FEF25–

ARTICLE IN PRESS

75 showed distinct dose-response behavior. Small airway dysfunction was not directly analyzed.

Zar et al. [43] compared four spacer devices in 88 children with acute asthma, assessing spirometry before and after bronchodilator administration. Among children with moderate-to-severe obstruction, FEF25–75 showed numerically greater percentage improvement than FEV1 across all spacers (e.g., commercial spacer: FEF25–75 +46% vs. FEV1 +23.5%). Post-treatment % predicted values were not reported, but the larger relative changes in FEF25–75 suggest heightened small airway responsiveness to bronchodilator delivery.

Malka et al. [46] analyzed 66 children with acute asthma to examine exhaled nitric oxide (FeNO) and lung function across baseline, exacerbation, and post-treatment recovery. FEF25–75 declined significantly during exacerbation and improved following treatment, with a pattern suggesting numerically larger % changes than FEV1. While the study emphasized FeNO as a marker of inflammation, the trajectory of FEF25–75 across time points may indicate small airway involvement, though it was not directly assessed.

ARTICLE IN PRESS

Yilmaz et al. [49] followed 57 children after moderate/severe asthma exacerbation, tracking lung function recovery via serial spirometry. FEF25–75 was the last parameter to normalize compared to FEV1 and PEF, with a median recovery time of 14 days versus 7 days for the other parameters. At day 1, the mean predicted FEF25–75 was 48%, improving to 91% by day 28. Severe exacerbation and allergic rhinitis were significantly associated with delayed FEF25–75 recovery (>7 days; $p = 0.006$ and $p = 0.003$, respectively). While small airway dysfunction was not directly analyzed, the prolonged recovery of FEF25–75 despite earlier normalization of FEV1

ARTICLE IN PRESS

and PEF suggests potential sensitivity to underlying small airway pathology, though this could alternatively reflect delayed resolution of inflammation or drug deposition limitations in the peripheral airways.

Comberiati et al. [50] evaluated spirometric changes in 92 children with asthma across three time points: baseline, exacerbation, and post-oral corticosteroid recovery. FEF25–75 declined during exacerbation and improved one week after treatment. Although small airway dysfunction was not a primary focus, the recovery pattern in FEF25–75 suggests small airway involvement during acute episodes.

Chandra et al. [51] conducted a randomised trial comparing terbutaline and salbutamol via MDI and spacer in 60 children with mild-to-moderate acute asthma. Both groups showed significant improvement in spirometry, including FEF25–75, with no significant differences between drugs. Although small airway dysfunction was not directly analyzed, the consistent improvement in FEF25–75 suggests similar bronchodilator effects in small airways across both treatments.

ARTICLE IN PRESS

Zanconato et al. [55] investigated the effects of a 5-day oral prednisone course in 30 children with asthma exacerbation. FEF25–75 improved significantly from 42% to 66% predicted ($p < 0.0005$), while FEV1 increased from 68% to 88% predicted ($p < 0.0005$). Although both parameters improved, the percentage increase was numerically greater for FEF25–75 (24%) than FEV1 (20%). However, no correlation was found between spirometric improvements and reductions in exhaled nitric oxide (FeNO) or exhaled carbon monoxide (ECO). These findings suggest that FEF25–75 may reflect peripheral airway recovery distinct from inflammatory marker trends.

Nuhoğlu et al. [58] randomised 60 children with mild-to-moderate asthma exacerbation to receive either high-dose inhaled budesonide or oral

ARTICLE IN PRESS

methylprednisolone plus medium-dose budesonide. Both groups showed significant improvement in FEF25–75 after treatment (Group I: 62.4 ± 25.5% to 84.2 ± 29.4%; Group II: 56.2 ± 25.2% to 74.0 ± 24.7%), but the between-group difference was not statistically significant (p = 0.819). A similar trend was observed in FEV1, which improved in both groups (Group I: 78.6 ± 18.2% to 92.3 ± 16.6%; Group II: 72.4 ± 17.0% to 85.7 ± 13.5%) with no significant between-group difference (p = 0.891). These findings suggest that both regimens improved lung function, including small airway indices, though the study did not specifically evaluate small airway dysfunction.

Lin et al. [59] compared terbutaline delivery via metered-dose inhaler (MDI) and nebulizer in 111 children with acute asthma. Both groups improved across spirometric parameters, including FEF25–75, but between-group differences were not statistically significant. FEF25–75 % increases were numerically greater in the MDI group (82.5% ± 298.4%) than in the nebulizer group (26.5% ± 48.3%), though this difference did not reach significance. The large standard deviation in the MDI group suggests considerable inter-individual variability in FEF25–75 response, which may limit its interpretability as a treatment outcome in this context.

Lin et al. [60] compared nebulized terbutaline with subcutaneous epinephrine in 90 children with acute asthma. Both treatments significantly improved FEV1 and FEF25–75, but the increase in FEF25–75 was significantly greater in the epinephrine group (65.7 ± 4.4% to 80.4 ± 5.0%) than in the terbutaline group (59.7 ± 4.3% to 65.3 ± 4.4%, p = 0.027). Among children with baseline FEV1 < 60% predicted, epinephrine yielded greater improvements in both FEF25–75 (p = 0.012) and oxygen saturation. Although small airway dysfunction was not explicitly analyzed, the disproportionate FEF25–75 response (particularly in patients with lower baseline FEV1)

ARTICLE IN PRESS

suggests that small airway indices may capture additional physiological changes not fully reflected in FEV1.

Studies with a similar response in FEF25–75 and FEV1

11 out of 35 studies reported similar levels of improvement in FEF25–75 and FEV1 during acute asthma exacerbations or recovery. Becker et al. [29] conducted a double-blind randomised trial comparing inhaled salbutamol with subcutaneous epinephrine in 40 children with acute asthma. Both groups showed significant within-group improvement in FEV1 and FEF25–75 after treatment. While small airway function was not analyzed separately, the observed FEF25–75 improvement indicates possible small airway involvement. However, spirometric values, including FEF25–75, were reported only in figure format, limiting extraction of precise numerical data.

Garra et al. [31] conducted a double-blind randomised trial comparing nebulized isoproterenol and metaproterenol in 30 hospitalized children and adolescents with severe acute asthma. Spirometry was performed at multiple intervals over 24 hours. Both FEV1 and FEF25–75 improved in both groups, with metaproterenol showing greater sustained increases. FEF25–75 showed significant between-group differences at select time points, suggesting potential sensitivity to treatment effects.

Scalabrin et al. [34] conducted a randomised controlled trial comparing three modes of salbutamol administration in 35 acute asthma episodes among 21 children. Spirometric assessment included both FEV1 and FEF25–75, measured over 8 hours. All groups showed improvement, with the most sustained bronchodilator effect observed with the closed nebulizer system. FEF25–75 changes paralleled FEV1 but were not analyzed separately.

ARTICLE IN PRESS

ARTICLE IN PRESS

Scalabrin et al. [35] conducted a randomised controlled trial comparing salbutamol, terbutaline, and fenoterol in 37 acute asthma episodes among 21 children. Spirometric outcomes included FEV1 and FEF25–75, measured repeatedly over 8 hours following nebulized treatment, although FEF25–75 was not analyzed independently. All three drugs produced significant bronchodilation with parallel improvements in FEV1 and FEF25–75.

Sharma et al. [36] conducted a diagnostic study in 50 children with acute asthma exacerbation to evaluate the correlation between single breath counting (SBC) and spirometry. Small airway function was not analyzed independently, however FEF25–75 showed a moderate and statistically significant correlation with SBC at baseline, though % change in FEF25–75 after treatment was not associated with change in SBC.

Lowenthal et al. [37] conducted a randomised controlled trial comparing aerosol delivery via facemask versus mouthpiece in 64 children with acute asthma. Spirometry was performed before and at multiple time points after treatment. Both FEV1 and FEF25–75 improved to a similar extent in both groups, with mean FEF25–75 increasing from 28.6% to 52.1% predicted in the facemask group and from 31.4% to 54.5% in the mouthpiece group, with no significant between-group differences.

Uden et al. [39] conducted a double-blind randomised trial comparing nebulized terbutaline with subcutaneous epinephrine in 19 children presenting with acute asthma. Spirometry was measured at baseline, 20 minutes, and 1-hour post-treatment. Both FEV1 and FEF25–75 improved similarly in both groups, with no significant difference between treatments. FEF25–75 remained markedly reduced post-treatment, suggesting persistent small airway obstruction despite clinical improvement. Small airway dysfunction was not explicitly analyzed.

ARTICLE IN PRESS

ARTICLE IN PRESS

Carter et al. [40] conducted a randomised, double-blind crossover trial in 11 children hospitalized with acute severe asthma to assess the short-term effects of heliox versus air. Spirometric outcomes included FEV1 and FEF25–75. Although both measures improved slightly, only FEF25–75 showed a statistically significant advantage with heliox: 32% ± 13% predicted on heliox vs. 29% ± 11% on air ($p = 0.006$). In contrast, FEV1 values were nearly identical between conditions (53% ± 18% vs. 52% ± 16%, $p = 0.36$), and no patient demonstrated a clinically meaningful difference exceeding 10% predicted. The magnitude of change in FEF25–75, while statistically significant, remained small and likely not clinically meaningful. Small airway dysfunction was not explicitly analyzed. One participant was 18 years old; this was retained in the analysis given the negligible impact on pediatric subgroup integrity.

Debley et al. [44] conducted a prospective observational study in 40 children hospitalized for acute asthma, assessing lung function and inflammatory biomarkers over 4 weeks. Spirometry was measured at enrolment and at 1-, 2-, and 4-weeks post-discharge. FEF25–75 improved rapidly post-hospitalization and plateaued by week 2, mirroring FEV1 trends. While small airway dysfunction was not directly analyzed, the trajectory of FEF25–75 recovery paralleled overall clinical improvement.

La Rosa et al. [48] conducted a randomised double-blind trial comparing high-dose inhaled flunisolide plus salbutamol to salbutamol alone in 28 children with acute asthma. Both groups demonstrated significant improvements in FEV1, PEF, and FEF25–75 over 15 days, with no significant differences between treatments and with FEF25–75 improvement mirroring trends seen in FEV1, suggesting small airway responsiveness during recovery.

ARTICLE IN PRESS

ARTICLE IN PRESS

Cengizlier et al. [52] compared subcutaneous glucagon to inhaled salbutamol in 40 children with acute asthma in a randomised trial. Both groups showed significant improvement in FEV1 and FEF25–75 at 30- and 120-minute post-treatment, with no statistically significant differences between the two treatments. Although small airway function was not explicitly discussed, the consistent improvement in FEF25–75 alongside FEV1 suggests comparable bronchodilator efficacy of glucagon and salbutamol across both large and small airways.

Studies with a lesser response in FEF25–75 compared with FEV1

3 out of 35 studies showed a lesser response as measured by tests of small airways function compared to large airways function. Caffarelli et al. [33] evaluated exhaled breath condensate (EBC) pH and spirometry in 12 children with asthma exacerbation before and after treatment. Small airway function was not a focus of analysis. While FEV1 showed statistically significant improvement, FEF25–75 increased only modestly and without significance. The authors found no correlation between EBC pH and spirometric values.

ARTICLE IN PRESS

Samransamruajkit et al. [53] conducted a randomised trial evaluating the effect of Flutter chest therapy in 40 children hospitalized with acute asthma. While FEV1 and FVC improved significantly more in the Flutter group, FEF25–75 also increased to a greater extent (mean 37.9% vs. 21.6% from baseline) but the between-group difference did not reach statistical significance ($p = 0.08$). This nonsignificant result may reflect greater variability in FEF25–75 response or baseline imbalances in disease severity. The study did not report responder analyses, but the observed FEF25–75 trajectory may suggest peripheral airway responsiveness during recovery.

ARTICLE IN PRESS

Lee et al. [56] compared spirometry and serum eosinophil cationic protein (sECP) in 22 children with asthma and allergic rhinitis during exacerbation and again 2 months later. FEV1 improved significantly from 80.9 ± 19.7% to 101.6 ± 10.5% predicted ($p = 0.002$), while FEF25–75 increased more modestly from 78.6 ± 26.7% to 92.8 ± 25.4% predicted ($p = 0.118$). The relatively weaker and non-significant change in FEF25–75 compared to FEV1 may reflect limited reversibility in the peripheral airways, slower resolution of inflammation, or insufficient treatment duration to normalize small airway function within the 2-month interval.

Cross-sectional studies without treatment comparisons

Yuan et al. [45] compared pulmonary function in 140 children divided into acute asthma, asthma remission, and cough variant asthma (CVA) groups, alongside 30 healthy controls. Small airway indices such as FEF25–75 were significantly reduced during exacerbation (50% ± 17 in acute asthma vs. 97% ± 12 in controls) and remained subnormal in the CVA group (FEF25–75 = 79% ± 22), similar to the remission group.

ARTICLE IN PRESS

Impulse oscillometry (IOS) and forced oscillation technique (FOT)

The four studies reporting IOS/FOT indices consistently found elevated markers of peripheral airway resistance, such as R5–R20 and AX, during asthma exacerbations. These abnormalities were often detected despite normal or mildly impaired conventional spirometric indices, and in some cases, IOS changes persisted into clinical remission. IOS parameters showed moderate correlation with spirometry but offered distinct information, with AX frequently emerging as the strongest predictor of bronchodilator response. While none of the studies performed formal diagnostic comparisons between IOS and spirometry, the findings suggest that IOS may provide a

ARTICLE IN PRESS

more accessible or sensitive alternative for detecting small airway dysfunction during acute asthma, particularly when MBW or spirometry are not feasible.

Wawszczak et al. [47] assessed peripheral airway function using IOS and MBW in 42 children with asthma, including 20 with acute exacerbations. In the exacerbation group, R5–R20 and AX were significantly elevated compared to the stable asthma group (R5–R20: 0.18 vs. 0.12; AX: 1.4 vs. 0.75), despite no significant difference in R5 or X5. MBW revealed abnormal Scond in 100% and abnormal Sacin in 76% of children with acute asthma, compared to 22.7% and 27%, respectively, in the stable group. Although spirometry indices were impaired, they did not differ significantly between the acute and stable groups (e.g., FEV1: 89% vs. 96% predicted, $p = 0.07$), suggesting that IOS and MBW captured abnormalities not detected by spirometry. These findings support the utility of advanced physiological measures in identifying small airway involvement during acute asthma exacerbations.

ARTICLE IN PRESS

Batmaz et al. [54] compared impulse oscillometry (IOS) and spirometry in 35 children with acute asthma, 107 with stable asthma, and 103 healthy controls. MMEF was significantly reduced in acute asthma compared to stable asthma and controls. IOS indices (including AX and R5–R20) were significantly higher in the acute group (AX: 2.16 ± 2.02 vs. 1.79 ± 1.25 vs. 1.23 ± 1.02 kPa/L; R5–R20: 0.35 ± 0.18 vs. 0.24 ± 0.10 vs. 0.18 ± 0.11 $kPa \cdot L^{-1} \cdot s$, all $p < 0.001$), reflecting greater small airway impairment. AX emerged as the strongest predictor of bronchodilator response. Although all participants successfully performed both spirometry and IOS, the authors noted IOS may be advantageous in patients unable to complete forced expiratory maneuvers.

Liu et al. [57] evaluated small airway function using impulse oscillometry in 40 children with acute asthma, 23 in remission, and 25 healthy controls. R5–R20, X5, Fres,

ARTICLE IN PRESS

and Zrs were all significantly elevated during exacerbation and improved after treatment, though some parameters remained abnormal in the remission group. The study did not include spirometry, so no comparisons could be made between IOS and conventional lung function indices. Nonetheless, IOS captured changes across disease states, supporting its potential utility in monitoring small airway involvement.

Yu et al. [61] examined 166 preschool children divided into four cohorts: acute asthma (n=44), non-acute asthma (n=41), cough variant asthma (CVA; n=46), and healthy controls (n=35). Lung function was assessed using impulse oscillometry (IOS) only; spirometry was not performed, likely due to the young age of participants. R5–R20, a marker of small airway dysfunction, was elevated across all asthma groups compared to controls, with the highest values seen in the acute asthma cohort (mean 0.45 $kPa \cdot L^{-1} \cdot s$ vs. 0.08 in controls, $p<0.01$). While not focused specifically on small airway dysfunction, the observed IOS trends suggest peripheral airway involvement, particularly during exacerbation, and highlight the utility of IOS in preschool children where spirometry is often not feasible.

ARTICLE IN PRESS

Synthesis of results

Across the 35 included studies, small airway function during pediatric asthma exacerbations was assessed using spirometry-derived indices, predominantly FEF25–75, and oscillometry parameters such as R5–R20 and AX. One study also reported multiple-breath washout (MBW) indices. Collectively, these data confirm that small airway testing is feasible in acute clinical settings, including in younger children and those with moderate to severe symptoms. All oscillometry reports described high test completion, although few studies formally compared completion or missingness against spirometry, which limits firm conclusions about relative feasibility.

ARTICLE IN PRESS

At presentation, small airway dysfunction was frequently observed. In studies using FEF25–75, baseline values were consistently reduced, often more so than FEV1. The oscillometry studies similarly reported elevated R5–R20 and/or AX during exacerbations, and the single MBW study documented abnormal ventilation inhomogeneity, including LCI and related indices. These abnormalities were evident across symptom severities and in some cases occurred when FEV1 remained within normal limits, supporting early involvement of the distal airways during acute episodes.

With treatment, most spirometry cohorts reported improvement in both FEV1 and FEF25–75 following bronchodilators and corticosteroids; however, FEF25–75 commonly showed larger relative change or slower normalization, suggesting distinct recovery kinetics. Oscillometry parameters typically declined after treatment yet sometimes remained abnormal beyond clinical improvement, indicating residual disturbance not detected by FEV1. Only a minority of studies related small airway measures to patient-centered outcomes such as symptom scores, peak flow, or oxygen saturation, and these analyses were generally exploratory and unadjusted for potential confounders.

ARTICLE IN PRESS

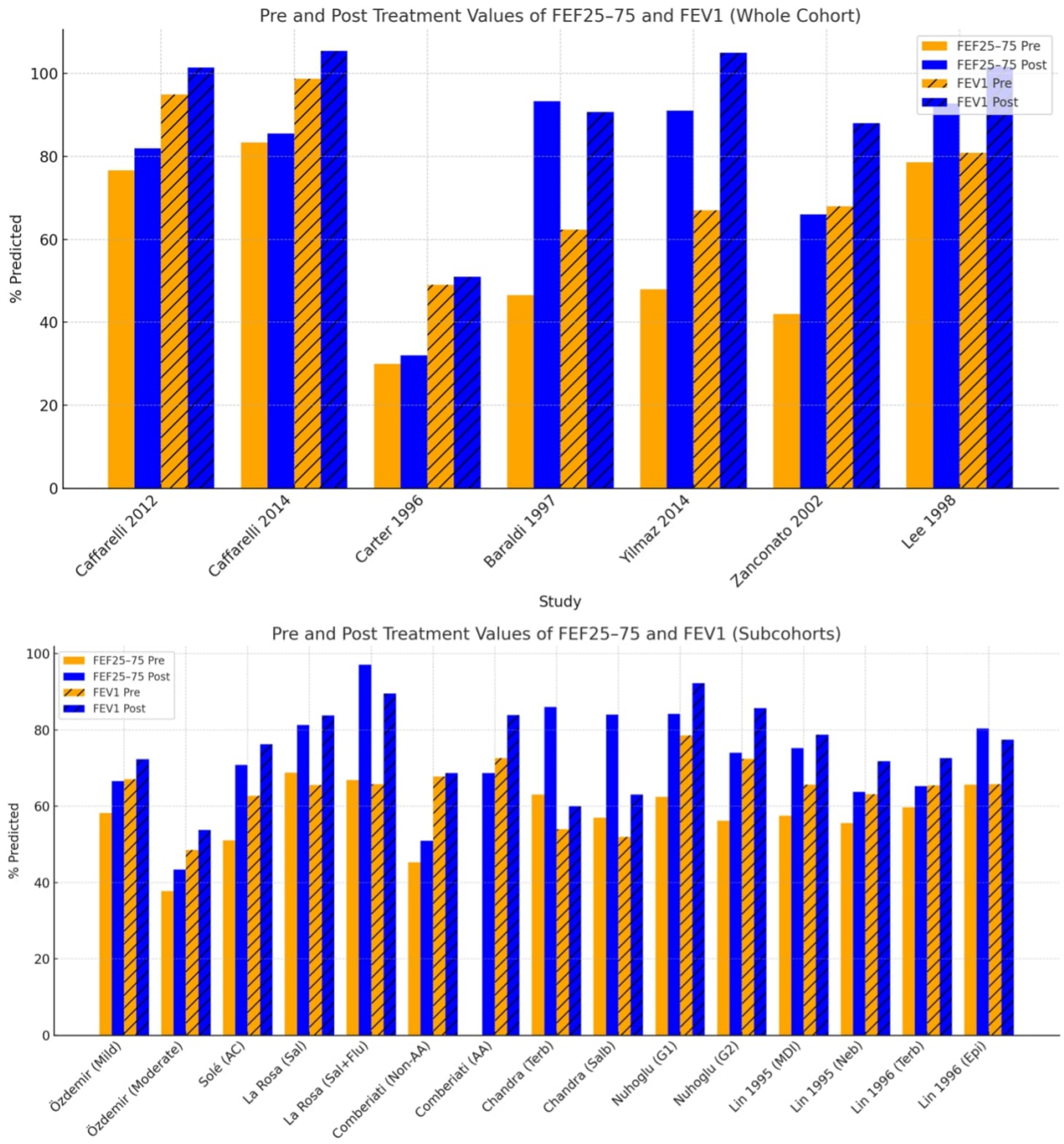


Figure 3. Comparison of FEV1 and FEF25–75 during paediatric asthma exacerbations.

(A) Whole-cohort pre- and post-treatment values for both FEV1 and FEF25–75. (B) Stratified subgroup findings (e.g., treatment arms or phenotypes). All values are expressed as % predicted.

ARTICLE IN PRESS

When appraising modalities with a view to clinical translation, FEF25–75 has the broadest evidence base because it is reported alongside routine spirometry in most studies. It is frequently impaired at onset, often shows larger acute change than FEV1, and may remain abnormal during early recovery. Its strengths are ubiquity and immediate availability, while limitations include effort dependence, variability, and inconsistent reporting and reference standards, particularly when treated as a secondary outcome without prespecified hypotheses. Oscillometry has a smaller but internally consistent literature. It is effort independent and appears highly feasible during acute illness and in preschool children, and parameters such as R5–R20 and AX identify abnormalities that may be missed by FEV1. The main constraints are fewer studies, device and algorithm heterogeneity, and limited linkage to hard clinical outcomes. MBW evidence during acute exacerbations is very limited, although ventilation inhomogeneity indices appear sensitive in principle; at present, its role in the acute setting is primarily research oriented.

ARTICLE IN PRESS

In practical terms, oscillometry appears the most suitable candidate for validation in acute care and in preschool settings where forced maneuvers are difficult, with R5–R20 and AX emerging as the leading indices for study. In school-aged children where spirometry is feasible, FEF25–75 is the most immediately accessible adjunct to FEV1 and may help identify residual or early small airway impairment, provided reporting is standardized and thresholds for clinically meaningful change are defined. MBW may contribute mechanistic insight but currently lacks an evidence base for routine application during exacerbations.

Overall, small airway tests are feasible and responsive during pediatric exacerbations and often reveal abnormalities not captured by FEV1 alone. However,

ARTICLE IN PRESS

726 heterogeneity in severity definitions, protocols, timing, reference equations, and
727 outcome reporting, together with frequent treatment of small airway indices as
728 secondary outcomes, limits certainty and precluded pooling. No included study
729 evaluated whether incorporating these measures altered management or improved
730 outcomes, underscoring the need for prospective validation that links predefined small
731 airway thresholds to clinical decisions and endpoints.
732
733 To enhance clinical readability, Table 2 summarizes the key findings by small
734 airway test type, highlighting the direction of change during exacerbation, response to
735 treatment, feasibility, and potential clinical interpretation. This overview complements
736 the detailed study-level data in Table 1 by providing a concise synthesis of patterns
737 relevant to clinical application.
738
739 Table 2. Summary of key findings by small airway test type during pediatric asthma
740 exacerbations

ARTICLE IN PRESS

Test type	No. of studies	Direction of change during exacerbation	Change following treatment	Comparison with FEV_1	Feasibility	Clinical interpretation
FEF25–75	31	Significantly reduced at presentation compared with baseline or controls	Improves after bronchodilator or corticosteroid therapy; often greater relative change than FEV_1	Frequently more sensitive to early or residual small airway dysfunction	High feasibility in school-aged children familiar with spirometry	Provides additional insight into distal airway involvement but is effort-dependent and variably reported
Impulse Oscillometry (IOS) (R5–R20, AX)	3	Elevated resistance and reactance during exacerbations	Decreases after treatment but may remain abnormal despite clinical recovery	Often detects abnormalities when FEV_1 is within normal limits	Excellent feasibility due to tidal-breathing technique; suitable for younger or acutely unwell children	Identifies peripheral airway involvement during acute illness; promising adjunct to spirometry
Multiple-Breath Washout (MBW) (LCI, Scond, Sacin)	1	Increased indices of ventilation inhomogeneity (e.g., LCI)	Partial improvement after treatment	Often abnormal despite normal FEV_1	Moderate feasibility; requires cooperation and time	Sensitive research tool for distal airway assessment; clinical utility during

ARTICLE IN PRESS

						exacerbations remains limited
Combined modalities	2	Limited data; concurrent impairment in both FEF25–75 and IOS parameters	Parallel improvement with treatment	Provide complementary information	Good	Preliminary evidence suggests that combining IOS and spirometry may enhance small airway assessment; requires validation

Abbreviations: IOS = impulse oscillometry; MBW = multiple-breath washout; LCI = lung clearance index; SAD = small airway dysfunction.

Discussion

To the authors' knowledge, this is the first systematic review to specifically examine whether physiological tests of small airway function can support the diagnosis or management of asthma exacerbations in children. We reviewed 35 studies involving children assessed during acute asthma episodes: 31 studies reported spirometry-derived FEF25–75, while four used impulse oscillometry (IOS), including one that also assessed multiple-breath washout (MBW) [47, 54, 57, 61].

Overall, the evidence demonstrates that small airway indices are frequently impaired during pediatric exacerbations and improve following treatment. Both FEF25–75 and IOS parameters (particularly R5–R20 and AX) consistently showed responsiveness to bronchodilators and corticosteroids, while MBW data remain limited. These findings indicate that small airway tests are feasible during acute illness and can reveal physiological changes not captured by FEV1 [27, 28, 30, 32, 38, 41-43, 46, 49-51, 55, 58-60]. Although small airway dysfunction (SAD) was not the primary focus in most studies, consistent response patterns in both FEF25–75 and IOS parameters (particularly

ARTICLE IN PRESS

R5–R20 and AX) suggest that these tests are feasible, responsive to treatment, and may offer added sensitivity in detecting physiological changes, especially in cases where FEV1 remains within normal limits [30, 47, 54, 57]. In practical terms, IOS appears particularly valuable in younger or acutely unwell children, where it is effort-independent and easier to perform, whereas in school-aged populations FEF25–75 remains the most accessible adjunct to FEV1 when consistently reported [15].

Effort-independent measures such as IOS appear particularly suitable in younger children and in those with moderate to severe exacerbations, where forced spirometry may be unreliable. All IOS studies achieved high completion rates during acute episodes and identified peripheral airway abnormalities not reflected in FEV1 [47, 54, 57, 61]. These findings suggest that IOS may be a feasible adjunct to spirometry in acute pediatric settings, although validation in larger cohorts is needed, particularly given its reliance on tidal breathing and high test completion rates [47]. Comparison with adult literature further supports these observations. In adults, small airway indices such as R5–R20 and FEF25–75 also deteriorate during exacerbations and recover more slowly than FEV1, as reported in the authors' previous systematic review [18]. However, feasibility appears greater in children, where tidal-breathing techniques like IOS achieve high completion even during acute illness. The persistence of small airway impairment after clinical recovery in both adults and children suggests a shared pattern of delayed peripheral recovery that may reflect inflammation, mucus retention, or ventilation inhomogeneity.

Several included studies demonstrated that FEF25–75 improved significantly after treatment, often outpacing FEV1 in relative or absolute gains [28, 30, 41, 50]. Conversely, other studies showed persistent suppression of FEF25–75 despite symptom

ARTICLE IN PRESS

resolution or FEV1 normalization, suggesting ongoing small airway impairment after clinical improvement [44, 49, 57]. This pattern implies that peripheral airway recovery may lag behind large airway improvement, which has implications for discharge criteria and post-exacerbation monitoring. These findings further emphasize that small airway parameters may capture residual physiological dysfunction beyond apparent symptom resolution, underscoring the need for longitudinal follow-up to determine whether delayed recovery predicts relapse or treatment failure.

Demographic or phenotypic differences in small airway recovery were also noted. Comberiati et al. observed slightly lower baseline and recovery values for FEF25–75 in African American children, although recovery trends were similar across racial groups [50]. Yilmaz et al. found that children with allergic rhinitis had delayed FEF25–75 recovery post-exacerbation, even when FEV1 had normalized, suggesting that comorbid atopy may prolong small airway dysfunction [49].

ARTICLE IN PRESS

In one IOS study, R5–R20 varied across asthma phenotypes, with highest values in acute asthma, intermediate values in cough variant and non-acute asthma, and the lowest in healthy controls [61]. While this suggests that IOS may differentiate airway resistance patterns across clinical subtypes, replication in larger studies is needed. Batmaz et al. proposed that combining IOS and spirometric indices may provide a more robust approach to SAD detection than either alone [54].

Despite these encouraging signals, the strength of the evidence remains limited by methodological and reporting issues. Most included studies were not designed to evaluate SAD explicitly, and small airway indices were often secondary endpoints without prespecified hypotheses or statistical powering. This secondary reporting

ARTICLE IN PRESS

increases the potential for selective reporting bias, as small airway outcomes were rarely predefined or consistently analyzed across studies. Many studies lacked stratification by exacerbation severity or phenotype, and there was significant heterogeneity in asthma definitions, severity classifications, testing protocols, timing of assessments, reference equations, and outcome definitions, which precluded meta-analysis [5, 6, 21, 30, 44]. Moreover, z-scores were rarely used, further limiting interpretability across age, sex, and ethnic groups, where percent predicted values may not adequately standardize for developmental differences [63]. Taken together, these factors support an overall certainty of evidence that is low to moderate, and future studies should apply structured frameworks such as GRADE to formally evaluate evidence certainty in accordance with PRISMA recommendations [20]; certainty was narratively judged as moderate for feasibility outcomes (e.g., completion rates for IOS and FEF25–75), low to moderate for physiological responsiveness (changes following bronchodilator or corticosteroid treatment), and low for patient-centered outcomes such as symptom correlations or recovery trajectories. Certainty was primarily downgraded due to inconsistency, indirectness, and imprecision across studies, reflecting variation in study design quality, small sample sizes, and the secondary reporting of small airway endpoints in most included work.

Several older trials were at high risk of bias due to unclear randomization, lack of blinding, small sample sizes, or inadequate reporting [29, 31, 34, 37]. Even among more recent studies, many were modestly sized or lacked a comparator group, reducing confidence in effect estimates [28, 30, 32, 49]. These limitations underscore the need for larger, prospective studies specifically designed to evaluate SAD during pediatric exacerbations using validated, harmonized protocols [38, 40]. Such designs should incorporate predefined small-airway endpoints, uniform reference standards, and

ARTICLE IN PRESS

clinically meaningful thresholds for change to facilitate translation into pediatric practice.

Nonetheless, these findings align with broader evidence that implicates SAD in asthma heterogeneity and poor control [16, 64]. In stable pediatric asthma, SAD is associated with higher symptom burden, inflammation, and exacerbation risk, even in children with normal FEV1 [12, 16]. This review extends those insights to acute care, indicating that small airway metrics may detect physiological abnormalities during exacerbations that are not captured by conventional spirometry. However, no included studies assessed whether use of these metrics altered clinical management or improved outcomes such as hospital admission, relapse, or length of stay.

If validated, small airway tests such as FEF25–75 and R5–R20 could complement traditional spirometry in clinical practice and research. These indices may help identify children with subclinical dysfunction or delayed physiological recovery [44, 49], inform post-discharge monitoring strategies [44, 57], and serve as objective tools for evaluating treatment response when symptoms or peak flow readings are unreliable particularly in young children unable to perform forced maneuvers [30, 35]. However, at present, their role should be considered exploratory, pending definition of reproducible thresholds for clinically meaningful change and formal evaluation of their impact on management outcomes.

Key gaps remain. No studies to date have established clinically meaningful thresholds for change in FEF25–75 or R5–R20 during exacerbations. Few examined correlations between small airway indices and patient-centered outcomes such as symptom scores, oxygenation, or PEF, and when reported, these were exploratory

ARTICLE IN PRESS

ARTICLE IN PRESS

without adjustment for confounders such as age, treatment intensity, or baseline severity. Longitudinal studies are especially needed, as several reports noted that small airway impairment can persist for weeks after symptom resolution, yet few studies followed participants beyond 4 weeks [44, 49, 57]. Investigating whether persistent SAD predicts future exacerbations or treatment failure could help define its role as a prognostic marker.

Limitations of this review include reliance on narrative synthesis due to study heterogeneity, which limited quantitative pooling, and inconsistent reporting in several studies, which constrained deeper data exploration. The findings should therefore be interpreted with caution, particularly as small airway indices were often secondary outcomes not prespecified or uniformly analyzed across studies. Potential publication bias cannot be excluded, as negative or inconclusive results may have been underreported. Moreover, differences in device algorithms, reference equations, and study timing reduce generalizability across populations. Nonetheless, the review provides the most comprehensive synthesis to date of small airway assessment during pediatric exacerbations and highlights key areas for standardization in future research.

ARTICLE IN PRESS

Conclusion

Physiological tests for small airway function appear feasible during paediatric asthma exacerbations and often show measurable changes compared to baseline or healthy controls. Several studies demonstrated that indices such as FEF25–75 and R5–R20 may respond earlier or more sensitively than FEV1 during treatment and recovery. However, the absence of standardised reference ranges, variable reporting quality, and heterogeneity in measurement protocols limit current clinical applicability. Importantly,

ARTICLE IN PRESS

no included studies evaluated whether incorporating small airway indices altered management decisions or patient outcomes such as admission or relapse. Further research is needed to evaluate small airway function specifically in the early phase of paediatric exacerbations and to determine whether these indices add prognostic or therapeutic value. Larger studies are also needed to assess the influence of age, phenotype, and comorbidities such as allergic rhinitis on small airway recovery. This review suggests these tests may offer added insight during asthma exacerbations, but they cannot yet be recommended for routine use without further validation.

List of abbreviations

Abbreviations: AX – reactance area; ATS – American Thoracic Society; CI – confidence interval; FEF25–75 – forced expiratory flow between 25% and 75% of vital capacity; FEV1 – forced expiratory volume in 1 second; FOT – forced oscillation technique; GINA – Global Initiative for Asthma; IOS – impulse oscillometry; IQR – interquartile range; LCI – lung clearance index; MBW – multiple-breath washout; NAEPP – National Asthma Education and Prevention Program; NOS – Newcastle–Ottawa Scale; PEF – peak expiratory flow; PRAM – Paediatric Respiratory Assessment Measure; PRISMA – Preferred Reporting Items for Systematic Reviews and Meta-Analyses; QUADAS-2 – Quality Assessment of Diagnostic Accuracy Studies 2; RCT – randomised controlled trial; RoB 2.0 – Risk of Bias 2.0; ROBINS-I – Risk Of Bias In Non-randomised Studies of Interventions; SAD – small airway dysfunction; SD – standard deviation; SEM – standard error of the mean.

Declarations

Ethics approval and consent to participate

ARTICLE IN PRESS

ARTICLE IN PRESS

Not applicable. Ethical approval was not required for this study as it involved the analysis of publicly available data from previously published studies.

Consent for publication

Not applicable. No individual person's data is included in this manuscript.

Availability of data and materials

All data supporting the findings of this study are available within the article and its Additional files. The systematic review protocol is publicly available in the PROSPERO repository (CRD42025623062, https://www.crd.york.ac.uk/PROSPERO/view/CRD42025623062).

Competing interests

The authors declare that they have no competing interests.

Funding

This work was supported by King Saud University, Riyadh, Saudi Arabia, under the sponsorship of the Government of Saudi Arabia through the Saudi Arabian Cultural Bureau in the UK.

ARTICLE IN PRESS

Authors' contributions

A.A. (Abdulrahman Alshehri) and E.S. (Elizabeth Sapey) conceived the study, developed the protocol, led the analysis, and contributed substantially to manuscript writing. A.A. and E.S. conducted abstract screening, full-text screening, and risk of bias assessment. A.A. and M.A. (Mohammed Almeshari) developed and executed the search strategy and contributed to data synthesis. A.A., M.I.A. (Mohammed Ibrahim Alshahrani), and R.A.S. (Robert Andrew Stockley) supported data interpretation and manuscript drafting. All authors reviewed and approved the final manuscript. A.A. is the guarantor of the study.

ARTICLE IN PRESS

Acknowledgements

Not applicable.

References

1. Global Initiative for Asthma (GINA). Global Strategy for Asthma Management and Prevention. 2025. Available from: https://ginasthma.org/gina-reports/. Accessed 12 Sep 2025.
2. Comberiati P, Cottini M, Landi M, Berti A, Lombardi C, Peroni D. Impulse oscillometry for the evaluation and management of pediatric asthma. Explor Asthma Allergy. 2023;1:219–29. doi:10.37349/eaa.2023.00022.
3. Asher MI, García-Marcos L, Pearce NE, Strachan DP. Trends in worldwide asthma prevalence. Eur Respir J. 2020;56(6):2002094. doi:10.1183/13993003.02094-2020. PMID: 32972987.
4. Asher I, Pearce N. Global burden of asthma among children. Int J Tuberc Lung Dis. 2014;18(11):1269-78.10.5588/ijtld.14.0170
5. Bloom CI, Nissen F, Douglas IJ, et al. Exacerbation risk and characterisation of the UK's asthma population from infants to old age. Thorax. 2018;73(4):313-20.10.1136/thoraxjnl-2017-210650
6. Bacharier L, Boner A, Carlsen KH, et al. Diagnosis and treatment of asthma in childhood: a PRACTALL consensus report. Allergy. 2008;63(1):5-34.10.1111/j.1398-9995.2007.01586.x
7. King GG, Bates J, Berger KI, Calverley P, de Melo PL, Dellacà RL, et al. Technical standards for respiratory oscillometry. Eur Respir J. 2020;55(2):1900753. doi:10.1183/13993003.00753-2019.
8. Castro-Rodriguez JA, Forno E, Rodriguez-Martinez CE, et al. Risk and protective factors for childhood asthma: what is the evidence? J Allergy Clin Immunol Pract. 2016;4(6):1111-22.10.1016/j.jaip.2016.05.003
9. Alblooshi A, Alkalbani A, Albadi G, et al. Is forced oscillation technique the next respiratory function test of choice in childhood asthma? World J Methodol. 2017;7(4):129–38.10.5662/wjm.v7.i4.129
10. Ducharme FM, Chalut D, Plotnick L, et al. The Pediatric Respiratory Assessment Measure: a valid clinical score for assessing acute asthma severity from toddlers to teenagers. J Pediatr. 2008;152(4):476-80.e1.10.1016/j.jpeds.2007.08.034
11. Komarow HD, Myles IA, Uzzaman A, et al. Impulse oscillometry in the evaluation of diseases of the airways in children. Ann Allergy Asthma Immunol. 2011;106(3):191-9.10.1016/j.anai.2010.11.011
12. Shi Y, Aledia AS, Galant SP, et al. Peripheral airway impairment measured by oscillometry predicts loss of asthma control in children. J Allergy Clin Immunol. 2013;131(3):718-23.10.1016/j.jaci.2012.09.022

ARTICLE IN PRESS

ARTICLE IN PRESS

13. Beydon N. Pulmonary function testing in young children. Paediatr Respir Rev. 2009;10(4):208-13.10.1016/j.prrv.2009.03.001
14. Robinson PD, Latzin P, Verbanck S, Hall GL, Horsley A, Gappa M, Thamrin C, Arets HGM, Aurora P, Fuchs SI, King GG, Lum S, Macleod K, Paiva M, Pillow JJ, Ranganathan S, Ratjen F, Singer F, Sonnappa S, Stocks J, Subbarao P, Thompson BR, Gustafsson PM. Consensus statement for inert gas washout measurement using multiple- and single-breath tests. Eur Respir J. 2013 Mar;41(3):507-22. doi:10.1183/09031936.00069712.
15. Beydon N, Davis SD, Lombardi E, et al. An official American Thoracic Society/European Respiratory Society statement: pulmonary function testing in preschool children. Am J Respir Crit Care Med. 2007;175(12):1304-45.10.1164/rccm.200605-642ST
16. Shi Y, Aledia AS, Tatavoosian AV, et al. Relating small airways to asthma control by using impulse oscillometry in children. J Allergy Clin Immunol. 2012;129(3):671-8.10.1016/j.jaci.2011.11.002
17. Schulze J, Biedebach S, Christmann M, et al. Impulse oscillometry as a predictor of asthma exacerbations in young children. Respiration. 2016;91(2):107-14.10.1159/000442448
18. Alshehri A, Alshahrani MI, Sapey E, Stockley RA, Almeshari M. Can measures of small airway dysfunction aid with the diagnosis or management of asthma exacerbations? A systematic review. BMJ Open Respir Res. 2025;12:e002926. doi:10.1136/bmjresp-2024-002926. PMID: 40571430; PMCID: PMC12207187.
19. National Institute for Health Research. PROSPERO: International prospective register of systematic reviews. CRD42025623062. Available from: https://www.crd.york.ac.uk/PROSPERO/display_record.php?ID=CRD42025623062. Accessed 12 Sep 2025.
20. Liberati A, Altman DG, Tetzlaff J, Mulrow C, Gøtzsche PC, Ioannidis JPA, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. J Clin Epidemiol. 2009;62(10):e1–34. doi:10.1016/j.jclinepi.2009.06.006. Epub 2009 Jul 23. PMID: 19631507.
21. Yi L, Zhao Y, Guo Z, et al. The role of small airway function parameters in preschool asthmatic children. BMC Pulm Med. 2023;23(1):219.10.1186/s12890-023-02515-3
22. Yee N, Markovic D, Buhr RG, et al. Significance of FEV3/FEV6 in recognition of early airway disease in smokers at risk of development of COPD: analysis of the SPIROMICS cohort. Chest. 2022;161(4):949-59.10.1016/j.chest.2021.10.046
23. Sterne JAC, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. BMJ. 2019;366:l4898. doi:10.1136/bmj.l4898. PMID: 31462531.
24. Sterne JAC, Hernán MA, Reeves BC, Savović J, Berkman ND, Viswanathan M, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ. 2016;355:i4919. doi:10.1136/bmj.i4919. PMID: 27733354; PMCID: PMC5062054.
25. Whiting PF, Rutjes AW, Westwood ME, et al. QUADAS-2: a revised tool for the quality assessment of diagnostic accuracy studies. Ann Intern Med. 2011;155(8):529-36.10.7326/0003-4819-155-8-201110180-00009
26. Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. Eur J Epidemiol. 2010;25:603-5.10.1007/s10654-010-9491-z

ARTICLE IN PRESS

ARTICLE IN PRESS

27. Nakanishi AK, Lamb BM, Foster C, et al. Ultrasonic nebulization of albuterol is no more effective than jet nebulization for the treatment of acute asthma in children. Chest. 1997;111(6):1505-8.10.1378/chest.111.6.1505
28. Özdemir A, Doğruel D. Efficacy of magnesium sulfate treatment in children with acute asthma. Med Princ Pract. 2020;29(3):292-8.10.1159/000506595
29. Becker AB, Nelson NA, Simons FER. Inhaled salbutamol (albuterol) vs injected epinephrine in the treatment of acute asthma in children. J Pediatr. 1983;102(3):465-9.10.1016/S0022-3476(83)80679-9
30. Chakraborti A, Lodha R, Pandey RM, Kabra SK. Randomized controlled trial of ipratropium bromide and salbutamol versus salbutamol alone in children with acute exacerbation of asthma. Indian J Pediatr. 2006;73(11):979–83. doi:10.1007/BF02758300. PMID: 17127777.
31. Garra B, Shapiro GG, Dorsett CS, et al. A double-blind evaluation of the use of nebulized metaproterenol and isoproterenol in hospitalized asthmatic children and adolescents. J Allergy Clin Immunol. 1977;60(1):63-8.10.1016/0091-6749(77)90084-7
32. Caffarelli C, Calcinai E, Rinaldi L, et al. Hydrogen peroxide in exhaled breath condensate in asthmatic children during acute exacerbation and after treatment. Respiration. 2012;84(4):291-8.10.1159/000341969
33. Caffarelli C, Dascola CP, Peroni D, Ricò S, Stringari G, Varini M, et al. Airway acidification in childhood asthma exacerbations. Allergy Asthma Proc. 2014;35(3):236–43. 10.2500/aap.2014.35.3740
34. Scalabrin DM, Naspitz CK. Efficacy and side effects of salbutamol in acute asthma in children: comparison of oral route and two different nebulizer systems. J Asthma. 1993;30(1):51–9. doi:10.3109/02770909309066380. PMID: 8428859.
35. Scalabrin DM, Solé D, Naspitz CK. Efficacy and side effects of beta2-agonists by inhaled route in acute asthma in children: comparison of salbutamol, terbutaline, and fenoterol. J Asthma. 1996;33(6):407-15.10.3109/02770909609068185
36. Sharma D, Saini AK, Jha DK, Tyagi A, Bisht SS. Single breath counting for diagnosis and prediction of severity of acute asthma exacerbation in children. Indian J Pediatr. 2023;90(8):776–80. doi:10.1007/s12098-022-04154-8. Epub 2022 Jun 1. PMID: 35648308.
37. Lowenthal D, Kattan M. Facemasks versus mouthpieces for aerosol treatment of asthmatic children. Pediatr Pulmonol. 1992;14(3):192-6.10.1002/ppul.1950140309
38. Sole D, Komatsu M, Carvalho K, et al. Pulse oximetry in the evaluation of the severity of acute asthma and/or wheezing in children. J Asthma. 1999;36(4):327-33.10.3109/02770909909068225
39. Uden DL, Goetz DR, Kohen DP, Fifield GC. Comparison of nebulized terbutaline and subcutaneous epinephrine in the treatment of acute asthma. Ann Emerg Med. 1985;14(3):229–32. doi:10.1016/S0196-0644(85)80445-5. PMID: 3883857.
40. Carter LER, Webb CCR, Moffitt CDR. Evaluation of heliox in children hospitalized with acute severe asthma: a randomized crossover trial. Chest. 1996;109(5):1256-61.10.1378/chest.109.5.1256
41. Baraldi E, Azzolin NM, Zanconato S, Dario C, Zacchello F. Corticosteroids decrease exhaled nitric oxide in children with acute asthma. J Pediatr. 1997;131(3):381–5. doi:10.1016/S0022-3476(97)80062-5. PMID: 9329413.
42. Shapiro GG, Furukawa CT, Pierson WE, et al. Double-blind, dose-response study of metaproterenol inhalant solution in children with acute asthma. J Allergy Clin Immunol. 1987;79(2):378-86.10.1016/0091-6749(87)90159-x
43. Zar H, Brown G, Donson H, et al. Home-made spacers for bronchodilator therapy in children with acute asthma: a randomised trial. Lancet. 1999;354(9183):979-82.10.1016/S0140-6736(98)12445-5

ARTICLE IN PRESS

ARTICLE IN PRESS

44. Debley JS, Cochrane ES, Redding GJ, Carter ER. Lung function and biomarkers of airway inflammation during and after hospitalization for acute exacerbations of childhood asthma associated with viral respiratory symptoms. Ann Allergy Asthma Immunol. 2012;109(2):114–20.e2. doi:10.1016/j.anai.2012.06.004. Epub 2012 Jun 26. PMID: 22840252; PMCID: PMC3430518.
45. Yuan J, An S-H, Gao W-J, et al. Comparative analysis of conventional pulmonary function test results in children with asthma or cough variant asthma. Zhongguo Dang Dai Er Ke Za Zhi. 2013;15(3):171-4.10.7499/j.issn.1008-8830.2013.03.002
46. Malka J, Covar R, Faino A, et al. The effect of viral infection on exhaled nitric oxide in children with acute asthma exacerbations. J Allergy Clin Immunol Pract. 2015;3(6):913-9.10.1016/j.jaip.2015.05.029
47. Wawszczak M, Kulus M, Peradzyńska J. Peripheral airways involvement in children with asthma exacerbation. Clin Respir J. 2022;16(2):97-104.10.1111/crj.13456
48. La Rosa M, Ranno C, Mandara G, et al. Double-blind study of inhaled salbutamol versus salbutamol plus high-dose flunisolide in exacerbation of bronchial asthma: a pilot study. Pediatr Asthma Allergy Immunol. 1997;11(1):23-30.10.1089/pai.1997.11.23
49. Yilmaz O, Bakirtas A, Ertoy Karagol H, et al. Allergic rhinitis may impact the recovery of pulmonary function tests after moderate/severe asthma exacerbation in children. Allergy. 2014;69(5):652-7.10.1111/all.12391
50. Comberiati P, Peroni D, Malka-Rais J, et al. Fractional exhaled nitric oxide response to oral corticosteroids in children with mild-to-moderate asthma: influence of race. Ann Allergy Asthma Immunol. 2020;125(4):440-6.e1.10.1016/j.anai.2020.06.036
51. Chandra P, Paliwal L, Lodha R, Kabra SK. Comparison of terbutaline and salbutamol inhalation in children with mild or moderate acute exacerbation of asthma. Indian J Pediatr. 2004;71(11):961–3. doi:10.1007/BF02828104. PMID: 15572812.
52. Cengizlier R, Çalışkol Y, Vidinlisan S, Önal Z. Glucagon in the treatment of acute asthma in children. Pediatr Asthma Allergy Immunol. 1999;13(4):177-80.10.1089/pai.1999.13.177
53. Samransamruajkit R, Chin TW, Yuengsrigul A, et al. Possible beneficial effect of chest physical therapy in hospitalized asthmatic children. Pediatr Asthma Allergy Immunol. 2003;16(4):295-303.10.1089/088318703322751345
54. Batmaz SB, Kuyucu S, Arıkoglu T, et al. Impulse oscillometry in acute and stable asthmatic children: a comparison with spirometry. J Asthma. 2016;53(2):179-86.10.3109/02770903.2015.1081699
55. Zanconato S, Scollo M, Zaramella C, et al. Exhaled carbon monoxide levels after a course of oral prednisone in children with asthma exacerbation. J Allergy Clin Immunol. 2002;109(3):440-5.10.1067/mai.2002.121954
56. Lee W-Y, Yang M-H, Chen L-C, Lin S-J, Huang J-L. Role of serum eosinophil cationic protein in the monitoring of children with bronchial asthma and rhinitis. Pediatr Asthma Allergy Immunol. 1998;12(2):129–38.10.1089/pai.1998.12.129
57. Liu X-M, Che S-Y, Liu X-R, et al. Values of impulse oscillometry for pulmonary function and adhesion molecules in childhood asthma. Zhongguo Dang Dai Er Ke Za Zhi. 2007;9(5):415-8
58. Nuhoğlu Y, Bahçeciler NN, Barlan IB, et al. The effectiveness of high-dose inhaled budesonide therapy in the treatment of acute asthma exacerbations in children. Ann Allergy Asthma Immunol. 2001;86(3):318-22.10.1016/S1081-1206(10)63306-6

ARTICLE IN PRESS

ARTICLE IN PRESS

59. Lin Y-Z, Hsieh K-H. Metered dose inhaler and nebuliser in acute asthma. Arch Dis Child. 1995;72(3):214-8.10.1136/adc.72.3.214
60. Lin Y-Z, Hsieh K-H, Chang L-F, Chu C-Y. Terbutaline nebulization and epinephrine injection in treating acute asthmatic children. Pediatr Allergy Immunol. 1996;7(2):95–9. doi:10.1111/j.1399-3038.1996.tb00113.x. PMID: 8902860.
61. Yu Z-W, Xie J-J, Qian J, et al. Roles of allergen testing in vitro and impulse oscillometry for lung function measurements in children with cough variant asthma. Zhongguo Dang Dai Er Ke Za Zhi. 2011;13(7):554-7
62. McGuinness LA, Higgins JP. Risk-of-bias VISualization (robvis): an R package and Shiny web app for visualizing risk-of-bias assessments. Res Synth Methods. 2021;12(1):55-61.10.1002/jrsm.1411
63. Quanjer PH, Stanojevic S, Cole TJ, Baur X, Hall GL, Culver BH, et al. Multi-ethnic reference values for spirometry for the 3–95-yr age range: the global lung function 2012 equations. Eur Respir J. 2012;40(6):1324–43. doi:10.1183/09031936.00080312.
64. Usmani OS, Singh D, Spinola M, et al. The prevalence of small airways disease in adult asthma: a systematic literature review. Respir Med. 2016;116:19-27.10.1016/j.rmed.2016.05.006

Figure legends

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart of included studies.

Figure 2. Risk of bias assessments. (A) Observational studies assessed using the Newcastle–Ottawa Scale (NOS) [26]. (B) Randomised controlled trials assessed using the revised Cochrane RoB 2.0 tool [23]. (C) Non-randomised interventional studies assessed using the ROBINS-I tool [24]. (D) Diagnostic test accuracy study assessed using the QUADAS-2 tool [25].

Figure 3. Comparison of FEV1 and FEF25–75 during paediatric asthma exacerbations.

ARTICLE IN PRESS

ARTICLE IN PRESS

(A) Whole-cohort pre- and post-treatment values for both FEV1 and FEF25–75. (B) Stratified subgroup findings (e.g., treatment arms or phenotypes). All values are expressed as % predicted.

Tables

Table 1. Characteristics of Included Studies

Study ID	Sample (n)	Severity Classification	Age (years)	small airway function Baseline	small airway function follow-up	FEV1	PEF	Diagnosis criteria
Nakanishi et al 1997[27] USA Randomised controlled trial	113 asthma exacerbation Ultrasonic Nebulizer (UN) N: 67 vs Jet Nebulizer (JN) N:46	Moderately severe asthma	UN:10.1 ± 3.3; JN: 11.0 ± 3.1	NR	NR	UN: 36.1 ± 18.5%; JN: 38.1± 16.9%	NR	Previous diagnosis of asthma
Özdemir et al 2020[28] Turkey non-randomised interventional study	115 children with acute asthma; Mild asthma (n = 50) Moderate asthma (n = 65)	Mild (FEV1 60–75%), Moderate (FEV1 40–59%)	Mild: 10.30 ± 2.91; Moderate: 10.54 ± 3.11	Mild: 58.20 ± 12.24%; Moderate: 37.77 ± 11.37%	Mild: 66.57 ± 16.95%; Moderate: 43.41 ± 14.19%	Mild: 67.14 ± 4.99%; Moderate: 48.50 ± 6.81%	Mild: 65.49 ± 12.32%; Moderate: 47.56 ± 11.78%	Previous diagnosis of asthma
Becker et al 1983* [29] Canada Randomised controlled trial	Acute asthma 40 (20 salbutamol, 20 epinephrine)	Not Specified	Salbutamol: 10.6 ± 0.7; Epinephrine: 10.4 ± 0.7	Reported in figure only	Reported in figure only	Reported in figure only	NR	Previous diagnosis of asthma

ARTICLE IN PRESS

Chakraborti et al 2006[30] India Randomised controlled trial	Asthma exacerbations 60 (30 salbutamol & ipratropium, 30 salbutamol & placebo)	Mild to moderate exacerbation	Group 1: 106 ± 25.8 months; Group 2: 118 ± 25.2 months	Group 1: 80.1 ± 65.5%; Group 2: 84.5 ± 129.8%	Group 1: 121.0 ± 90.7%; Group 2: 72.2 ± 33.3%	Group 1: 72.3 ± 37.1%; Group 2: 57.4 ± 21.0%	Group 1: 64.36 ± 28.3%; Group 2: 57.3 ± 19.20%	Previous diagnosis of asthma
Garra et al 1997* [31] USA Randomised controlled trial	30 hospitalized children with acute asthma (15 received isoproterenol, 15 received metaproterenol)	Severe exacerbation requiring hospitalization	Isoproterenol: 10.0 ± 1.1; Metaproterenol: 10.1 ± 1.1	Isoproterenol: 15.3 ± 2.6%; Metaproterenol: 11.8 ± 1.6%	Reported in figure only	Isoproterenol: 30.3 ± 3.4%; Metaproterenol: 24.8 ± 4.4%	NR	Previous diagnosis of asthma
Caffarelli et al 2012[32] Italy non-randomised interventional study	12 healthy controls. 22 asthmatic children with acute asthma exacerbation.	Mild–moderate exacerbation	Healthy controls: 11.7 ± 2.8; Asthmatics: 9.4 ± 2.7	Healthy controls: 106.2 ± 19.8%. Asthmatics: 76.6 ± 18.3%	Asthmatics: 81.9 ± 19.3%	Healthy controls: 112.6 ± 15.8%; Asthmatics: 95.0 ± 11.9%	NR	Global Initiative for Asthma
Caffarelli et al 2014[33] Italy non-randomised interventional study	Healthy Control Group: 28 children Asthma Exacerbation Group: 12 children	Mild–moderate exacerbation	Healthy controls: 5.1 ± 1.4; Asthmatics: 9.3 ± 2.4	Asthmatics: 83.4 ± 12.8%	Asthmatics: 85.5 ± 22.17%	Asthmatics: 98.7 ± 0.66%	NR	Global Initiative for Asthma
Scalabrin et al 1993[34] Brazil Randomised controlled trial	35 episodes in 21 children; Oral salbutamol: 13, Open nebulization: 11, Closed-port nebulization: 11	Acute bronchial asthma	mean 10.1 years (range: 7–14 years)	OR:1.12 ± 0.44 L/s; ON:0.78 ± 0.34 L/s; CN: 0.79 ± 0.41 L/s	NR	OR: 55.27%; ON: 45.07%; CN: 42.44%	NR	Previous diagnosis of asthma
Scalabrin et al 1996[35] Brazil Randomised controlled trial	37 episodes in 21 children; groups: Salbutamol (11), Terbutaline (12), Fenoterol (14)	Acute bronchial asthma	Mean: 10.41 years (range 7–14)	SAL: 0.79 ± 0.41; TER: 0.71 ± 0.44; FEN: 1.01 ± 0.65	NR	SAL: 42.4%; TER: 42.9%; FEN: 53.1%	NR	Previous diagnosis of asthma
Sharma et al 2023[36]	50 children with acute asthma	Mild–moderate and	9.66 ± 1.63 years	1.24 ± 0.55 L/s	1.56 ± 0.64 L/s	1.09 ± 0.32 L/S	2.40 ± 0.96 L/S	Global Initiative

India Diagnostic test accuracy	exacerbation.	severe exacerbation						for Asthma
Lowenthal et al 1992[37] Randomised controlled trial	64 children with acute asthma; Facemask: 32, Mouthpiece: 32.	mild to moderate	Facemask: 13.5 ± 4.4; Mouthpiece: 13.0 ± 3.6	Facemask: 28.6 ± 14.3%; Mouthpiece: 31.4 ± 16.5%	NR	Facemask: 45.3 ± 16.3%; Mouthpiece: 50.7 ± 15.1%	Facemask: 53.4 ± 22.2%; Mouthpiece: 54.7 ± 20.0%	American Thoracic Society (ATS)
Sol et al 1999[38] Brazil Observational study	174 children with acute exacerbation (Allergy Clinic: 74, Emergency Room: 100)	mild to severe based on clinical score	AC: 8.6 y: ER: 4.6 y	AC group: 51.1%, ER: not reported	AC group: 70.8%, ER: not reported	AC group: 62.7%, ER: not reported	NR	Previous diagnosis of asthma
Uden et al 1985[39] USA Randomised controlled trial	Asthma exacerbation: total analyzed = 19 (Terbutaline: 7; Epinephrine: 12)	moderate severity	Mean: 11.2 ± 2.9 years (range: 5.8–17.5)	Reported in figure only	Terbutaline: 31.8 ± 18.6%; Epinephrine: 39.0 ± 12.2%	Reported in figure only	NR	Previous diagnosis of asthma
Carter et al 1996[40] USA Randomised controlled trial	11 children ACUTE ASTHMA	Acute severe asthma	Range: 7–18 years	30 ± 14%	32 ± 14%	49 ± 16%	48 ± 16%	American Thoracic Society (ATS)
Baraldi et al 1997* [41] Italy Non-randomised, controlled, interventional study	Total = 39 Acute asthma = 16 Healthy controls = 16 Stable asthma with albuterol challenge = 7	NR	Healthy controls: 9.3 ± 0.5 years; Stable asthma: 10.6 ± 0.7 years; Exacerbation group: 9.3 ± 0.5 years.	Healthy controls: 114.4 ± 4.9%; Asthmatics: 46.5 ± 5.1 %	Asthmatics: 93.3 ± 8.0%	Healthy controls: 93.8 ± 2.2 %; Asthmatics: 62.4 ± 4.4 %	NR	objective testing and clinical criteria.
Shapiro et al 1987* [42] USA Randomised controlled trial	100 children (ages 6–12) with acute asthma exacerbation; randomised to saline (n=25), metaproterenol 5 mg (n=25), 10 mg (n=25), or	Mild to moderate acute exacerbation	Age: 6–12 years (mean not reported)	Placebo: 0.50 ± 0.05 L/s; Metaproterenol 5 mg: 0.53 ± 0.06 L/s; 10 mg: 0.72 ± 0.07 L/s; 15 mg: 0.74 ± 0.07 L/s	Reported in figure only	Placebo: 0.77 ± 0.07 L; Metaproterenol 5 mg: 0.84 ± 0.08 L; 10 mg: 0.99 ± 0.07 L; 15 mg: 1.02 ± 0.08 L	Placebo: 1.16 ± 0.10 L/s; Metaproterenol 5 mg: 1.29 ± 0.12 L/s; 10 mg: 1.41 ± 0.10 L/s; 15 mg: 1.45 ± 0.11 L/s	Previous diagnosis of asthma

ARTICLE IN PRESS

	15 mg (n=25).							
Zar et al 1999[43] South Africa Randomised controlled trial**	88 children with acute asthma. Arms: Conventional spacer (n=22), sealed bottle (n=22), unsealed bottle (n=22), polystyrene cup (n=22).	PEF 60–79% (mild), PEF 20–59% (moderate/severe)	Conventional spacer: 10.3 (8.9–12.4) years; Sealed bottle: 9.2 (7.9–10.7) years; Unsealed bottle: 9.1 (7.9–10.7) years; Cup: 9.7 (7.5–11.3) years.	Conventional spacer: 37 (27–52); Sealed bottle: 32 (24–47); Unsealed bottle: 40 (26–50); Cup: 37.5 (27–51).	NR	Conventional spacer: 67 (57–77); Sealed bottle: 67.5 (51–74); Unsealed bottle: 64.5 (52–75); Cup: 65 (53–77).	Conventional spacer: 58.5 (45–69); Sealed bottle: 59.5 (43–68); Unsealed bottle: 59 (32–73); Cup: 54.5 (44–67).	Previous diagnosis of asthma
Debley 2012[44] USA Observational study	Asthmatic children (exacerbation group): 34 completed follow-ups; Healthy controls: 40 enrolled.	Severe exacerbation	Healthy controls: 10.8 ± 3.2 years; Asthma (exacerbation): 9.7 ± 2.9 years.	Healthy controls: 90.7 ± 19.7%; Asthma (exacerbation): 43.3 ± 26.7%	Not reported for the whole cohort	Healthy controls: 99.3 ± 10.8%; Asthma (exacerbation): 76.9 ± 20.5% (FEV1).	NR	National Heart, Lung, and Blood Institute/ National Asthma Education and Prevention Program
Yuan et al 2013[45] China Observational study	Total patients: 170 Healthy controls: 30 Stable asthma: 50 Acute asthma exacerbation: 50 Cough-variant asthma (CVA): 40	Mild to moderate exacerbation	Healthy: 8.6 ± 2.1; Stable asthma: 8.0 ± 2.1; Asthma exacerbation: 8.3 ± 2.0; CVA: 7.9 ± 2.0	Healthy: 97 ± 12%; Stable asthma: 84 ± 18%; Asthma exacerbation: 50 ± 17%; CVA: 79 ± 22%	NR	Healthy: 107 ± 16 %; Stable asthma: 98 ± 14%; Asthma exacerbation: 69 ± 12%; CVA: 98 ± 14%	Healthy: 106 ± 14 %; Stable asthma: 95 ± 12%; Asthma exacerbation: 67 ± 14%; CVA: 93 ± 17%	Previous diagnosis of asthma
Malka et al 2015[46] USA Observational study	66 children ACUTE ASTHMA (during stable period & exacerbation)	Mild, moderate, and severe persistent asthma.	11 ± 3.2 years (Mean ± SD)	stable period (Mean ± SD): 76.6 ± 26.8% predicted; Exacerbation (Mean ± SEM): 43 ± 2% predicted	71.0 ± 4.9% predicted (model-estimated mean from Table III)	Stable period (Mean ± SD): 95.0 ± 16.0% predicted; Exacerbation (Mean ± SEM): 72 ± 2% predicted	NR	Previous diagnosis of asthma
Maria Wawszczak [47] Poland**	42 children with asthma Groups: Stable	Mixed	Age (median, IQR): Stable: 12 (7–	R5–R20 ($kPa \cdot L^{-1} \cdot s$, median [IQR]): Stable:	NR	(z-score, median [IQR]): Stable asthma:	NR	Global Initiative for Asthma

Observational study	asthma (n = 22), Exacerbation (n = 20)		17); Exacerbation: 12.5 (6–17)	0.12 (0.07–0.16); Exacerbation: 0.18 (0.13–0.295) MBW z-scores (median [IQR]): • Scond: Stable: 0.35 (–0.5 to 1.5); Exacerbation: 7.7 (4.6 to 8.9) • Sacin: Stable: 0.4 (–1.3 to 1.7); Exacerbation: 2.2 (1.6 to 3.2) • LCI: Stable: 1.6 (0.7 to 3.4); Exacerbation: 8.2 (5.85 to 11.4)		0.0 (–1.37 to 0.59); Exacerbation: –1.98 (–3.01 to –1.42)		
La Rosa et al 1997[48] Italy Randomised controlled trial	28 children with acute asthma exacerbation: Salbutamol + Flunisolide (n=15), Salbutamol only (n=13)	Intermittent or mild persistent exacerbation	Age range: 5–14 years Mean age: 9.0 ± 2.5 years (whole cohort)	68.8 ± 7.6% (salbutamol), 66.9 ± 5.0% (salbutamol + flunisolide)	Day 15: 81.3 ± 5.5%(salbutamol), 97.1 ± 6.4% (salbutamol + flunisolide)	65.6 ± 2.8%(salbutamol), 65.8 ± 3.3% (salbutamol + flunisolide)	72.7 ± 4.7 %(salbutamol), 63.8 ± 3.8% (salbutamol + flunisolide)	Clinical and spirometry-based
Yilmaz et al 2014[49] Turkey Observational study	Total patients: 57 children with moderate to severe asthma exacerbation	GINA 2012 criteria (moderate /severe)	10.7 ± 2.4 years	48 ± 17%	Day 35: 90 ± 6%	67 ± 17%	63 ± 16%	Global Initiative for Asthma
Comberiati et al 2020[50] USA Observational study	92 children with asthma exacerbation: 37 African American, 55 Non-African American (White and Hispanic).	Mild to moderate persistent asthma (NHLBI)	Overall cohort: Mean age = 11.0 ± 3.3 years	African American – Stable: 71.4 ± 35.6%; Non-African American – Stable: 75.0 ± 24.2%; African	African American: 68.7 ± 33.6; Non-African American: 67.8 ± 21.9	African American – Stable: 86.3 ± 17.3%; Non-African American – Stable: 96.3 ± 14.2%	NR	Based on previous physician diagnosis and NAEPP treatment context

ARTICLE IN PRESS

ARTICLE IN PRESS

				American – Exacerbation: 51.0 ± 24.6%; Non-African American – Exacerbation: 45.3 ± 17.7%				
Chandra et al 2004† [51] India Randomised controlled trial	60 children with mild or moderate asthma exacerbation (Terbutaline: n = 31; Salbutamol: n = 29).	Mild and moderate (defined by symptoms and PEFR % predicted)	Terbutaline: Median age = 10 years (IQR 9–11); Salbutamol: Median age = 9 years (IQR 7.8–11)	Terbutaline: 63% (52.9–77); Salbutamol: 57% (52.7–66.3) (likely mislabeled as PEFR in Table 1)	Terbutaline: 86% (75.5–99); Salbutamol: 84% (67.5–98.1)	Terbutaline: 54% (45–62); Salbutamol: 52% (48–63.8) median (95% confidence intervals)	Terbutaline: 69% (59.5-75); Salbutamol: 72% (61.3-90.1)	Not reported
Cengizlier et al 1999[52] Turkey Randomised controlled trial	40 children with acute asthma exacerbation (Glucagon: n=20; Salbutamol: n=20)	NR	9.9 ± 2.1 (glucagon), 10.7 ± 2.1 (salbutamol)	Reported in figure only	Reported in figure only	Reported in figure only	Reported in figure only	American Thoracic Society (ATS)
Samransamruajkit et[53] al 2003 USA Randomised controlled trial	40 children with acute asthma exacerbation; Flutter therapy (n=20), Standard care (n=20)	Moderate asthma exacerbation	standard care: 10.82 ± 0.7; Flutter: 9.19 ± 0.42	standard care: 27.1 ± 4.1; Flutter: 20.6 ± 3.1	Reported in figure only	Standard care: 46.4 ± 3.5%; Flutter: 38.8 ± 3.0%	NR	Previous diagnosis of asthma
Batmaz et al 2015[54] Turkey Observational study	245 children total 35 with acute asthma exacerbation, 107 with stable asthma, and 103 healthy controls	Mild	Acute asthma: 10.82 ± 2.53 years; Stable asthma: 10.57 ± 2.5 years; Healthy controls: 11.35 ± 2.54 years	Acute asthma: FEF25–75 = 52.3 ± 28.4% predicted; R5–R20 = 0.35 ± 0.18 kPa·L^{-1}·s; Stable asthma: FEF25–75 = 70.6 ± 24.5% predicted; R5–R20 = 0.24 ± 0.10 kPa·L^{-1}·s; Healthy	NR	Acute asthma: 83.5 ± 19.5%; Stable asthma: 95.7 ± 13.2%; Healthy controls: 105.9 ± 13.9%	NR	National Heart, Lung, and Blood Institute

				controls: FEF25–75 = 100.7 ± 27.9% predicted; R5–R20 = 0.18 ± 0.11 kPa·L^{-1}·s.				
Zanconato et al 2002* [55] Italy Observational study	30 children with acute asthma exacerbation; 29 healthy controls	NR	Age (Mean ± SEM): Asthma: 10.5 ± 0.5 years; Healthy controls: 8.5 ± 0.5 years	Asthma: 42%	Asthma: 66%	Asthma: 68%	NR	American Thoracic Society (ATS)
Lee et al 1998[56] Taiwan Observational study	34 children (22 with asthma + allergic rhinitis, 12 with allergic rhinitis only)	NR	8 years (range 3y3m to 14y)	Asthma exacerbation: 78.6 ± 26.7%	Asthma exacerbation: 92.8 ± 25.4%	Asthma exacerbation: 80.9 ± 19.7%	NR	NHLBI 1997 guidelines
Liu et al 2007[57] China Observational study	Acute asthma (exacerbation): 40 patients; Remission (same cohort post-treatment): 23 patients; Healthy controls: 25 patients	Mild-to-moderate asthma exacerbation	Healthy controls: 4.9 ± 0.2 years; Acute asthma: 4.1 ± 0.3 years; Asthma remission (acute asthma after treatment): 4.5 ± 0.8 years	R5–R20 (kPa·L^{-1}·s, Healthy controls: 0.24 ± 0.06; Acute asthma: 0.53 ± 0.24;	R5–R20 (kPa·L^{-1}·s, Asthma remission (acute asthma after treatment): 0.35 ± 0.11	NR	NR	Chinese national guideline for pediatric asthma
Nuhoğlu et al 2001[58] Turkey Randomised controlled trial	Total: 60 children with acute asthma exacerbation Group I (high-dose inhaled budesonide): n=31; Group II (oral methylprednisolone + medium-dose budesonide): n=29	Mild-to-moderate acute asthma exacerbations	age: 8.91 ± 2.60 years; age: 9.54 ± 3.05 years	Group I: Baseline: 62.42 ± 25.50; Group II: Baseline: 56.17 ± 25.18	Group I: Post-treatment: 84.23 ± 29.44; Group II: Post-treatment: 74.03 ± 24.71	Group I: Baseline: 78.61 ± 18.23; Group II: Baseline: 72.38 ± 17.03	NR	American Thoracic Society (ATS)

ARTICLE IN PRESS

Lin et al 1995[59] Taiwan Randomised controlled trial	Total: 111 children with acute asthma exacerbation MDI + spacer (n=56), nebulized terbutaline (n=55)	Mild-to-moderate acute asthma	MDI Group n = 56, Age: 8.1 ± 2.6 years; Nebulizer Group n = 55, Age: 8.4 ± 3.0 years	Group 1 (MDI): Baseline: 57.53 ± 28.02; Group 2 (Nebulizer): Baseline: 55.63 ± 28.77	Group 1 (MDI): 75.24 ± 31.25; Group 2 (Nebulizer): 63.82 ± 28.83	Group 1 (MDI): Baseline: 65.74 ± 20.33; Group 2 (Nebulizer): Baseline: 63.14 ± 20.50	Group 1 (MDI): Baseline: 57.56 ± 19.60%; Group 2 (Nebulizer): Baseline: 60.30 ± 21.32%	American Thoracic Society (ATS)
Lin et al 1996*[60] Taiwan non-randomised interventional study	90 children with acute asthma exacerbation (Terbutaline nebulization: n = 45; Epinephrine injection: n = 45)	Mild, moderate, and severe based on FEV1 and clinical scoring	Group 1 (Terbutaline): n = 45, age: 7.78 ± 0.35 years; Group 2 (Epinephrine): n = 45, age: 7.47 ± 0.31 years	Group 1 (Terbutaline): Baseline: 59.74 ± 4.33; Group 2 (Epinephrine): Baseline: 65.73 ± 4.39	Group 1 (Terbutaline): Post-treatment: 65.34 ± 4.39; Group 2 (Epinephrine): Post-treatment: 80.36 ± 5.04	Group 1 (Terbutaline): Baseline: 65.45 ± 3.08; Group 2 (Epinephrine): Baseline: 65.81 ± 3.05	NR	American Thoracic Society (ATS)
Yu et al 2011[61] China Observational study**	166 preschool children (age 3–6 years): Acute asthma (n=44); non-acute asthma (n=41); Cough variant asthma (n=46); Healthy controls (n=35)	Mild-to-moderate	(reported range: 3–6 years; group-specific values not separately reported)	R5–R20 (kPa·L^{-1}·s, Median [IQR]): Healthy controls: 0.08 (0.04–0.17); Acute asthma: 0.20 (0.15–0.28); non-acute asthma: 0.13 (0.10–0.19); Cough variant asthma: 0.12 (0.08–0.16)	NR	NR	NR	Chinese national and international diagnostic standards

ARTICLE IN PRESS

Legend: Values reported in mean (SD)
*Values reported in mean (SEM)
**Values reported in Median (interquartile range, 25th–75th percentile)
† Median with 95% confidence intervals
ID = identification, NR = Not Reported.

ARTICLE IN PRESS

Table 2. Summary of key findings by small airway test type during pediatric asthma exacerbations

Test type	No. of studies	Direction of change during exacerbation	Change following treatment	Comparison with FEV_1	Feasibility	Clinical interpretation
FEF25–75	31	Significantly reduced at presentation compared with baseline or controls	Improves after bronchodilator or corticosteroid therapy; often greater relative change than FEV_1	Frequently more sensitive to early or residual small airway dysfunction	High feasibility in school-aged children familiar with spirometry	Provides additional insight into distal airway involvement but is effort-dependent and variably reported
Impulse Oscillometry (IOS) (R5–R20, AX)	3	Elevated resistance and reactance during exacerbations	Decreases after treatment but may remain abnormal despite clinical recovery	Often detects abnormalities when FEV_1 is within normal limits	Excellent feasibility due to tidal-breathing technique; suitable for younger or acutely unwell children	Identifies peripheral airway involvement during acute illness; promising adjunct to spirometry
Multiple-Breath Washout (MBW) (LCI, Scond, Sacin)	1	Increased indices of ventilation inhomogeneity (e.g., LCI)	Partial improvement after treatment	Often abnormal despite normal FEV_1	Moderate feasibility; requires cooperation and time	Sensitive research tool for distal airway assessment; clinical utility during exacerbations remains limited
Combined modalities	2	Limited data; concurrent impairment in both FEF25–75 and IOS parameters	Parallel improvement with treatment	Provide complementary information	Good	Preliminary evidence suggests that combining IOS and spirometry may enhance small airway assessment; requires validation

Abbreviations: IOS = impulse oscillometry; MBW = multiple-breath washout; LCI = lung clearance index; SAD = small airway dysfunction.

Additional files

- Additional file 1: PRISMA 2020 checklist
  - Format: PDF
  - Title of data: Completed PRISMA 2020 checklist
  - Description: Completed PRISMA 2020 checklist for this systematic review.

ARTICLE IN PRESS

- Additional file 2: PICO framework
  - Format: DOCX
  - Title of data: Predefined PICO framework
  - Description: Inclusion criteria for the systematic review structured according to a predefined PICO framework.

- Additional file 3: Search strategy
  - Format: DOCX
  - Title of data: Full electronic database search strategy

    Description: Detailed search strategies for Ovid Embase, Ovid MEDLINE, EBSCO CINAHL, and CENTRAL.

- Additional file 4: List of excluded studies
  - Format: DOCX
  - Title of data: Excluded studies with reasons

    Description: List of studies excluded at the full-text screening stage, with reasons for exclusion.

ARTICLE IN PRESS